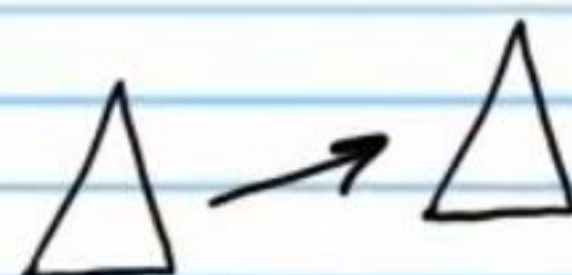


Transformations and the Library of Relations

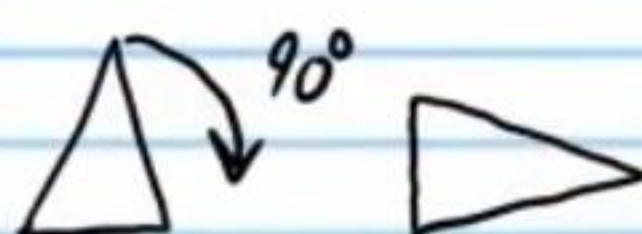
Review on Transformations

I) Translation.



II) Rotation.

* There must be a center of rotation



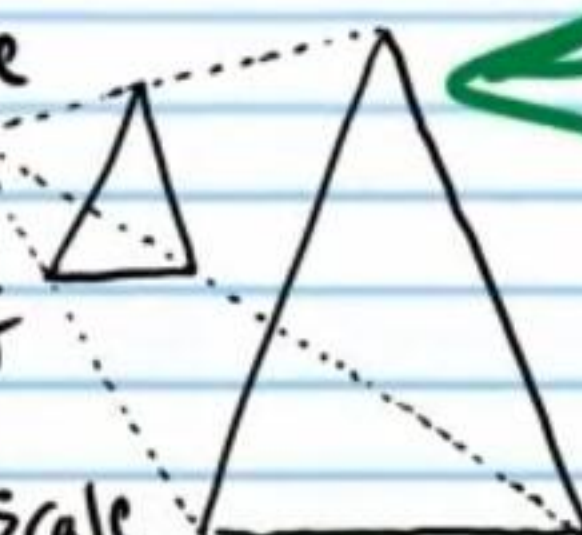
III) Reflection.

* There must be a line of reflection.



IV) Dilation.

* There must be a center of dilation and a scale factor.



No Change in Size.

Rigid

Size Changes.

Non-Rigid

Relations

Relation: A connection between two values (usually called x and y).

Three Ways to Represent a Relation

Examples

I. Equation

$$y = x^2$$

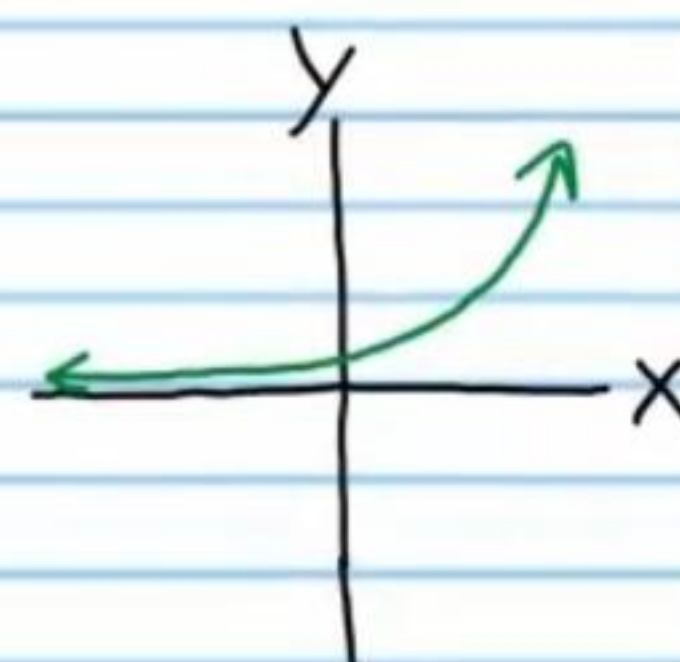
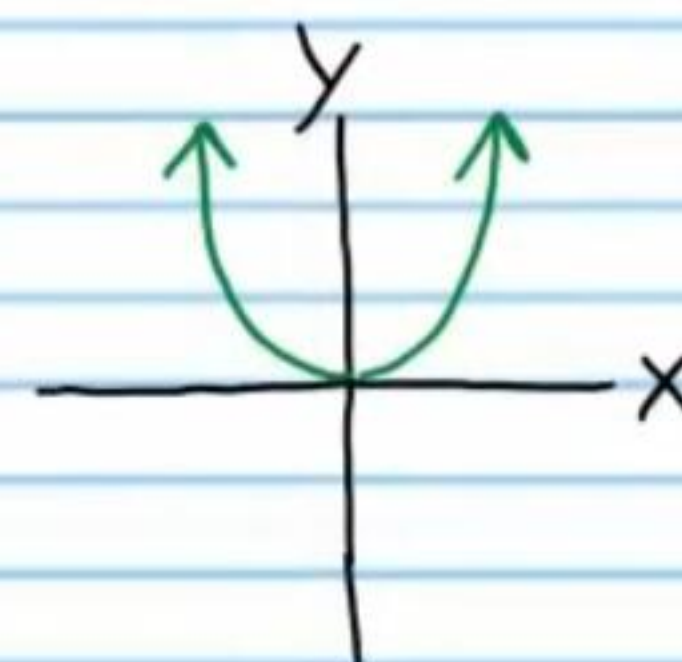
$$y = 3^x$$

II. Table

x	-2	-1	0	1	2
y	4	1	0	1	4

x	-2	-1	0	1	2
y	$\frac{1}{9}$	$\frac{1}{3}$	1	3	9

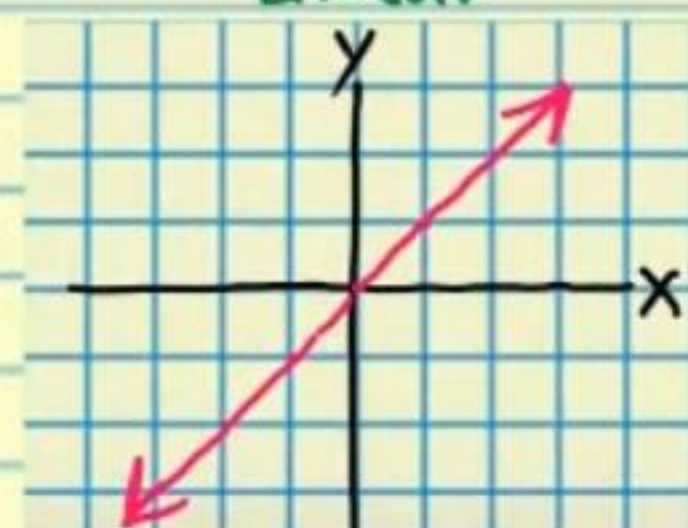
III. Graph



The Library of Relations

* Also called "parent" relations.

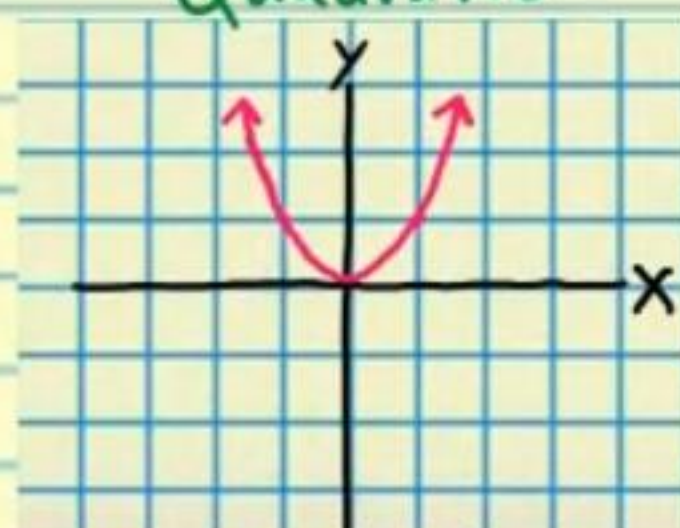
Linear



$$Y = X$$

X	-2	-1	0	1	2
Y	-2	-1	0	1	2

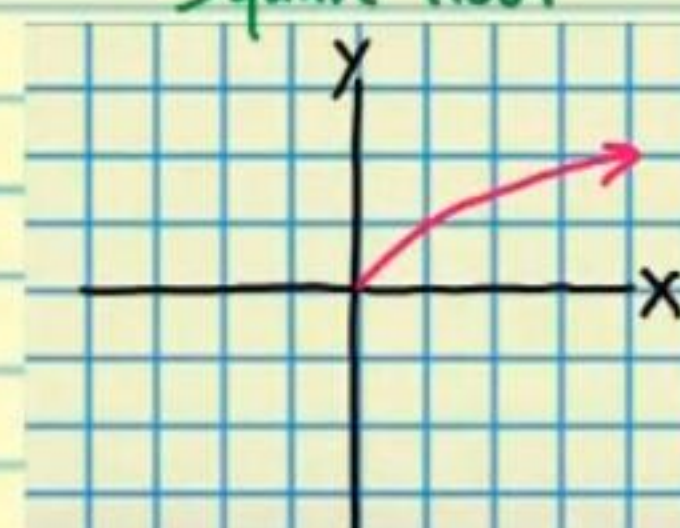
Quadratic



$$Y = x^2$$

X	-2	-1	0	1	2
Y	4	1	0	1	4

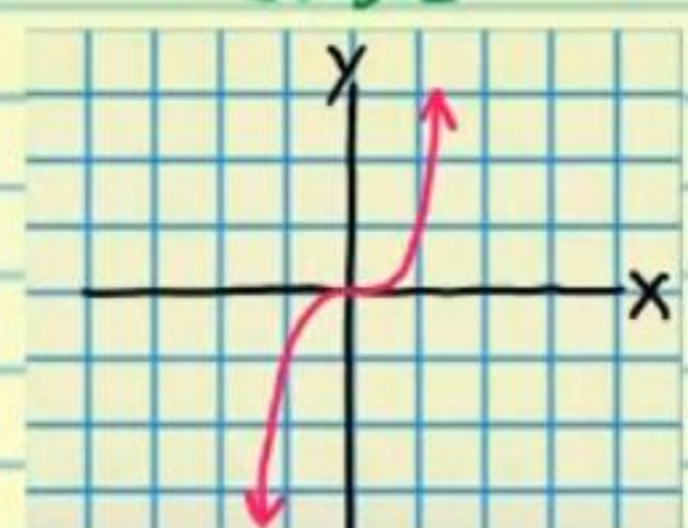
Square Root



$$Y = \sqrt{X}$$

X	0	1	4	9	16
Y	0	1	2	3	4

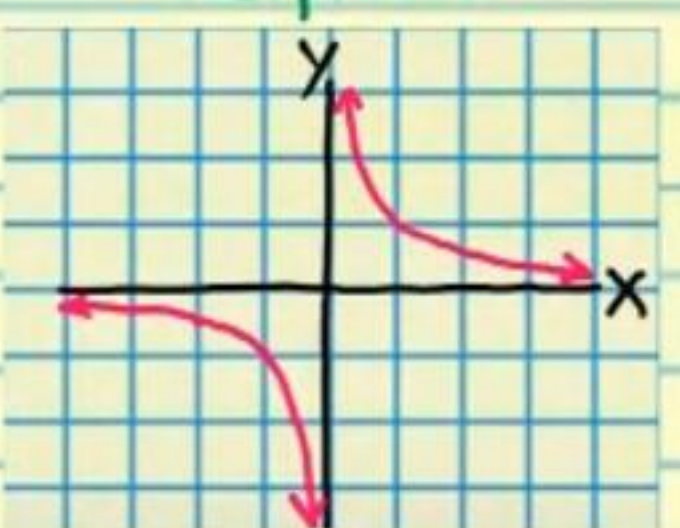
Cubic



$$Y = x^3$$

X	-2	-1	0	1	2
Y	-8	-1	0	1	8

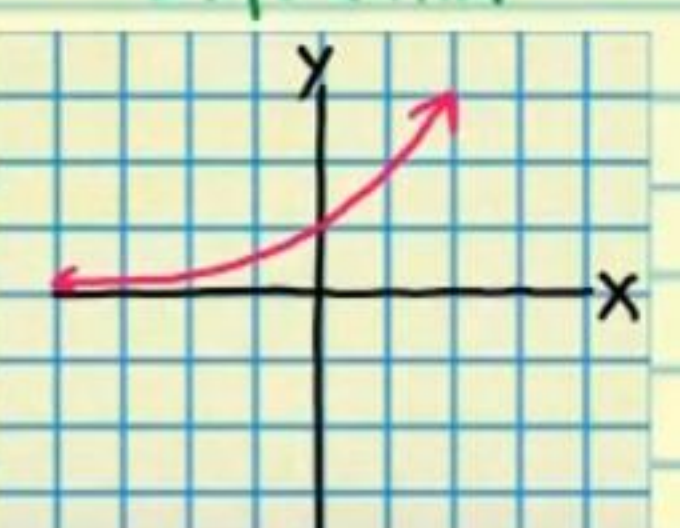
Reciprocal



$$Y = \frac{1}{x}$$

X	-2	-1	0	1	2
Y	-1/2	-1	und	1	1/2

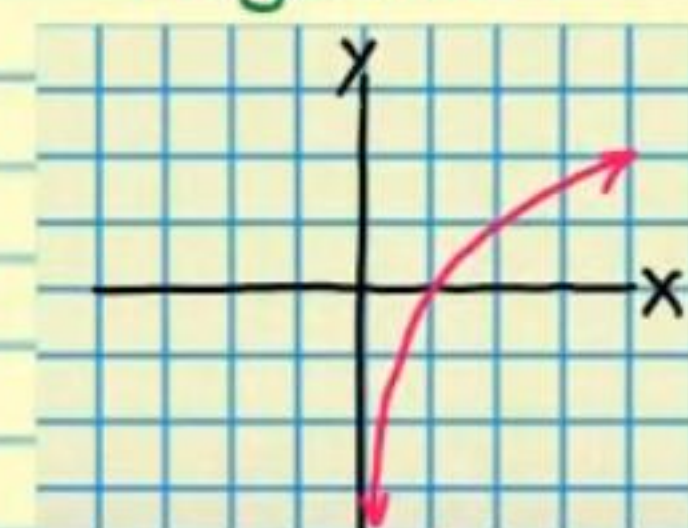
Exponential



$$Y = a^x, a > 1$$

X	-	-	0	-	-
Y	-	-	1	-	-

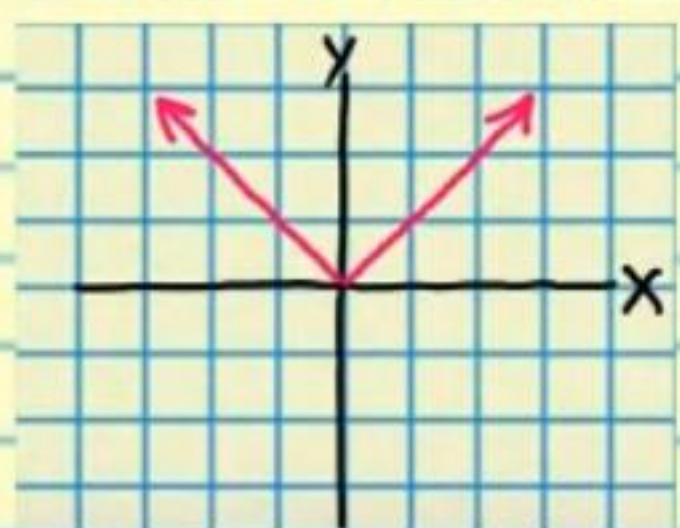
Logarithmic



$$Y = \log_a X$$

X	0	1	-	-	-
Y	und	0	-	-	-

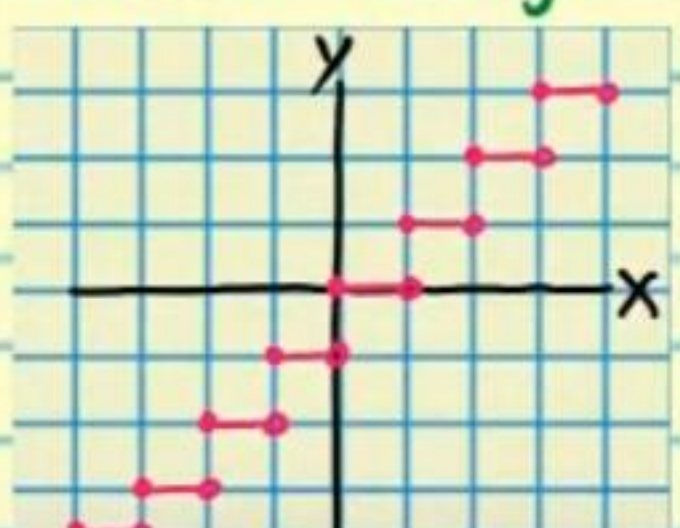
Absolute Value



$$Y = |x|$$

X	-2	-1	0	1	2
Y	2	1	0	1	2

Greatest Integer



$$Y = [x]$$

X	-2 1/2	-1	0	1/2	1 1/2
Y	-3	-1	0	0	1

Translations

Translating a Graph Vertically

If y is changed to $y - k$ in an equation, then its corresponding graph is moved up k units (assuming that k is positive).

If y is changed to $y + k$ in an equation, then its corresponding graph is moved down k units (assuming that k is positive).

Translating a Graph Horizontally

If x is changed to $x - k$ in an equation, then its corresponding graph is moved to the right k units (assuming that k is positive).

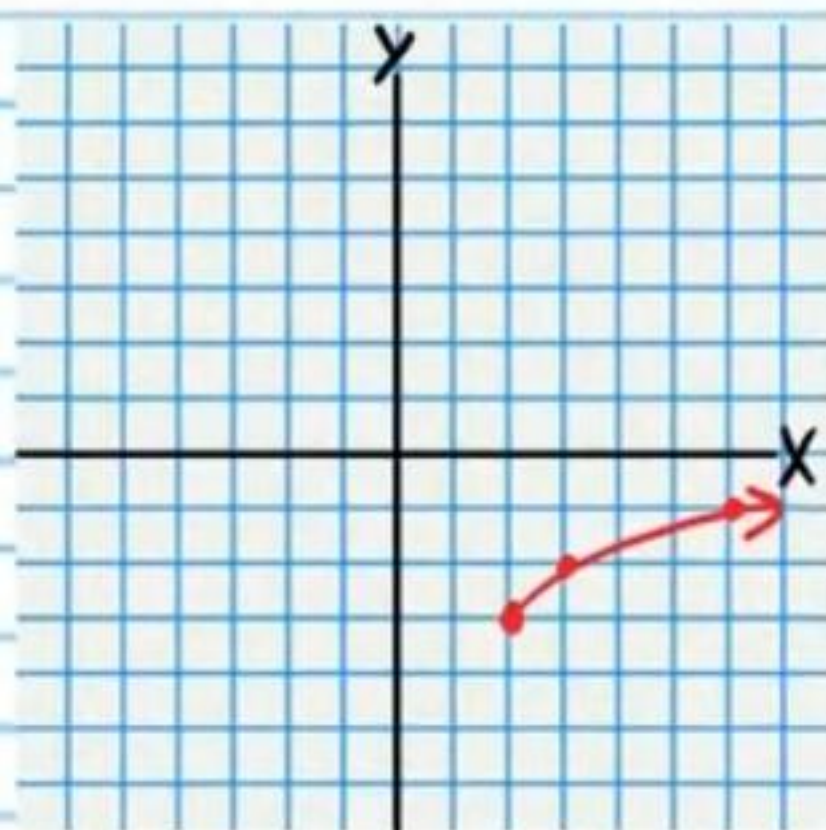
If x is changed to $x + k$ in an equation, then its corresponding graph is moved to the left k units (assuming that k is positive).

Write the parent relation and the parent equation of each relation below and use this information to graph the modified equation.

① $y+3 = \sqrt{x-2}$

Parent Relation: Square Root

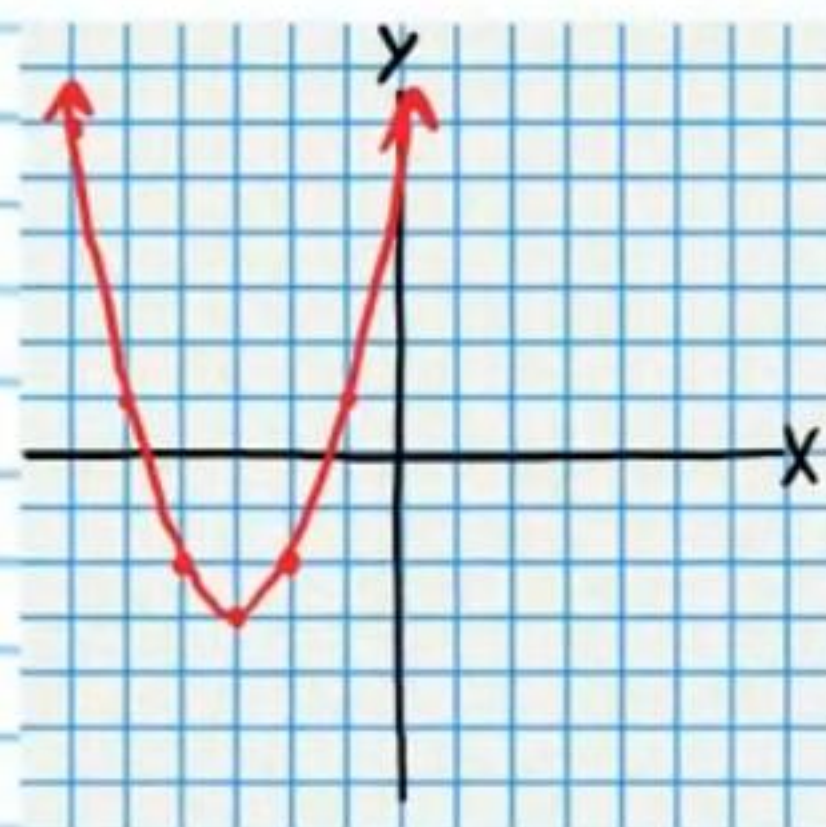
Parent Equation: $y = \sqrt{x}$



② $y+3 = (x+3)^2$

Parent Relation: Quadratic

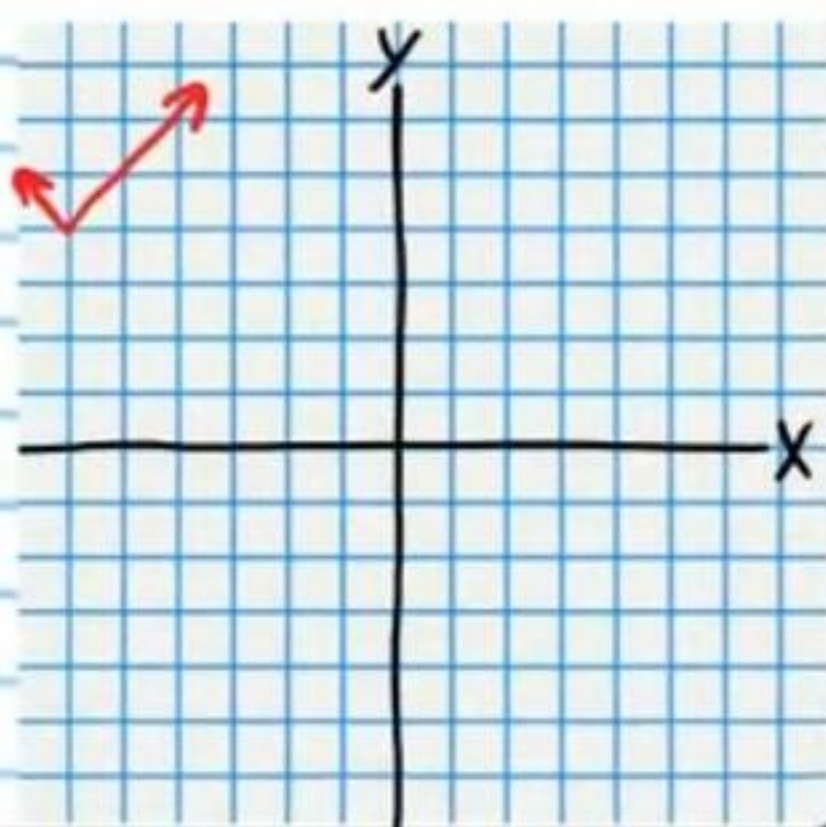
Parent Equation: $y = x^2$



③ $y-4 = |x+6|$

Parent Relation: Absolute Value

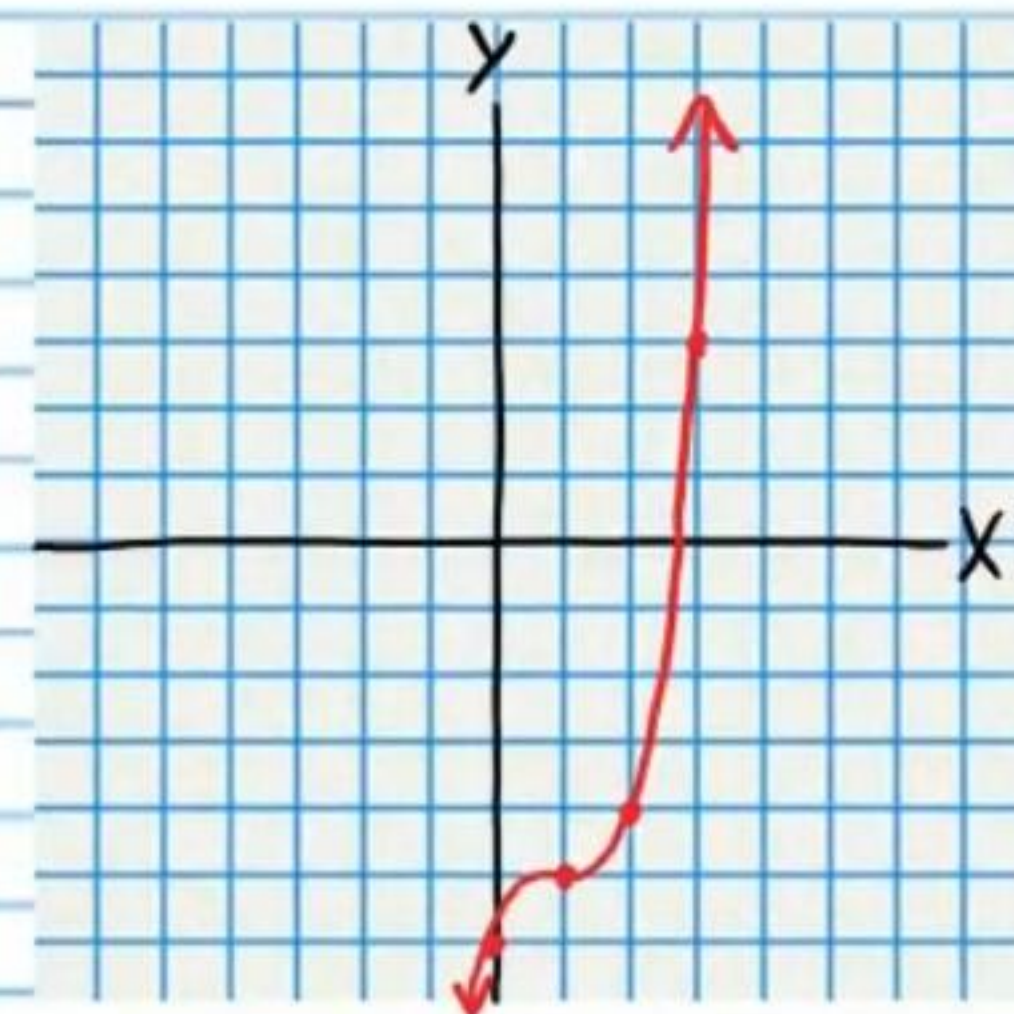
Parent Equation: $y = |x|$



④ $y+5 = (x-1)^3$

Parent Relation: Cubic

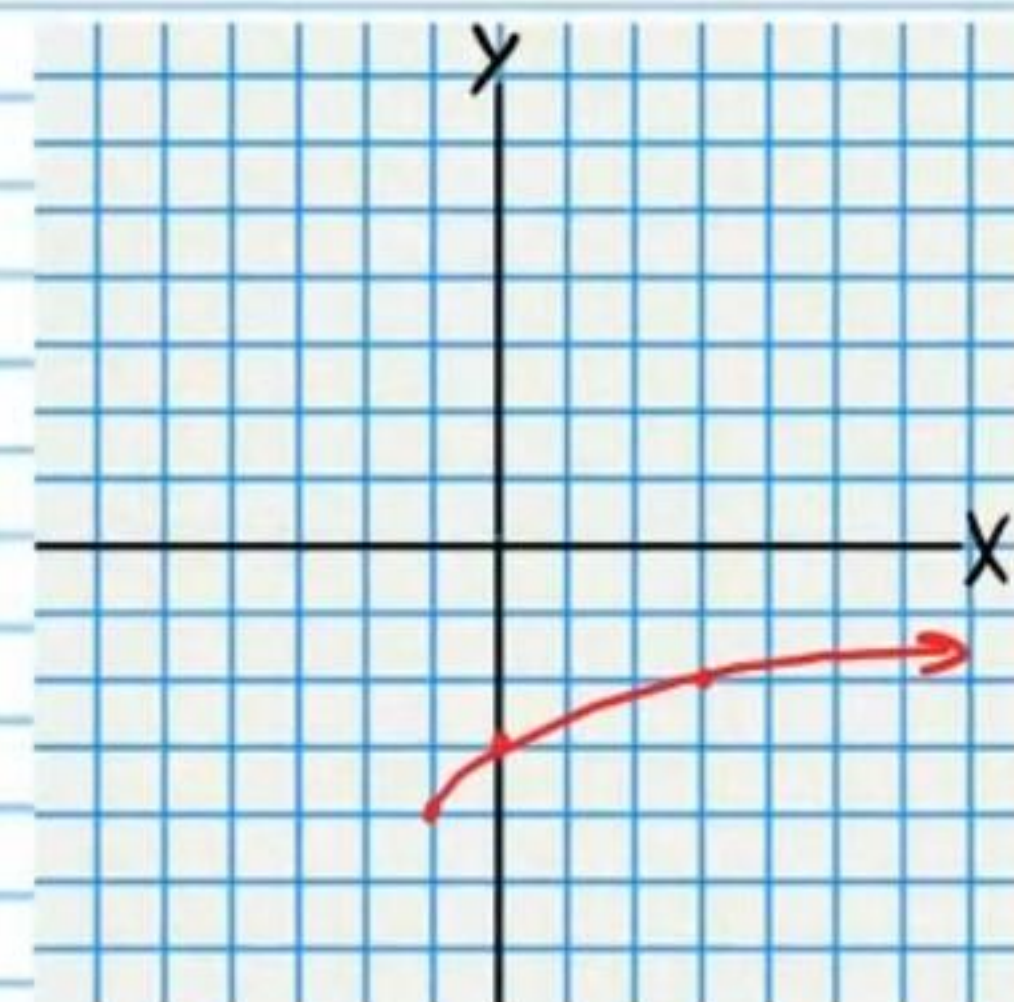
Parent Equation: $y = x^3$



⑤ $y+4 = \sqrt{x+1}$

Parent Relation: Square Root

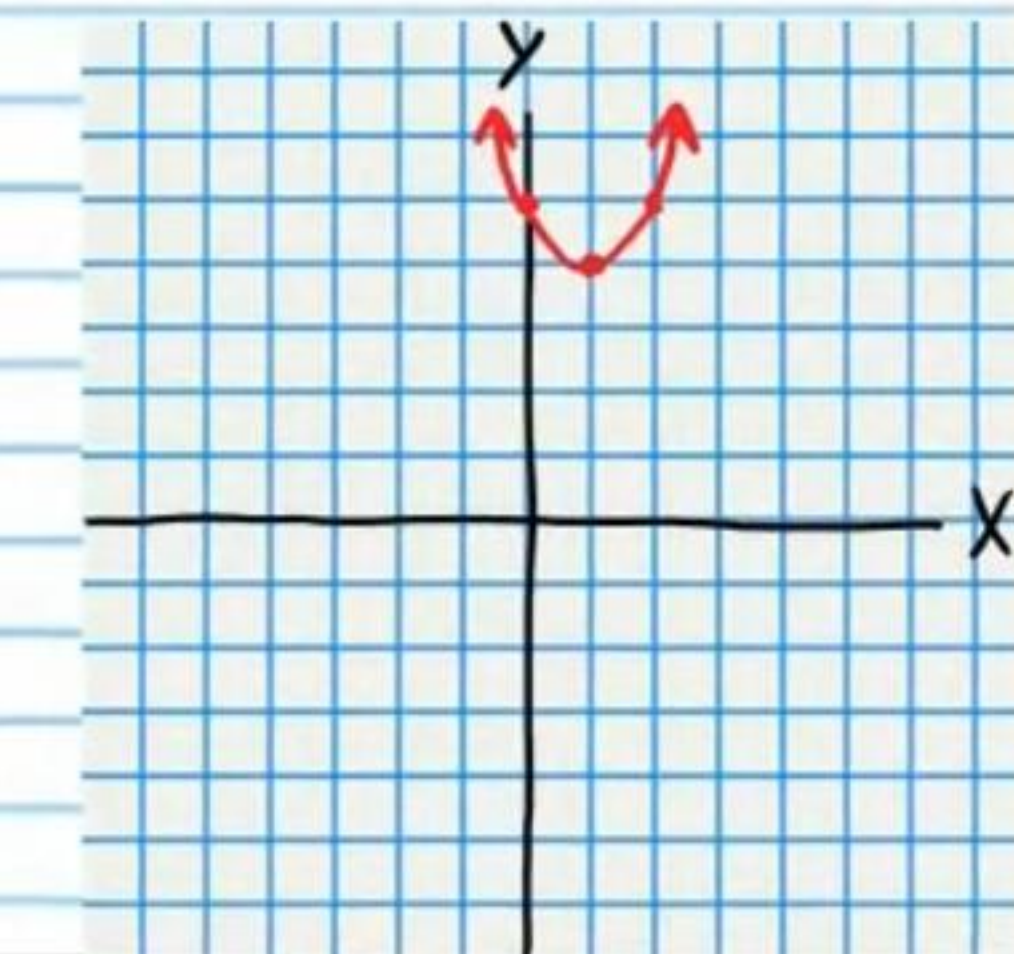
Parent Equation: $y = \sqrt{x}$



⑥ $y-4 = (x-1)^2$

Parent Relation: Quadratic

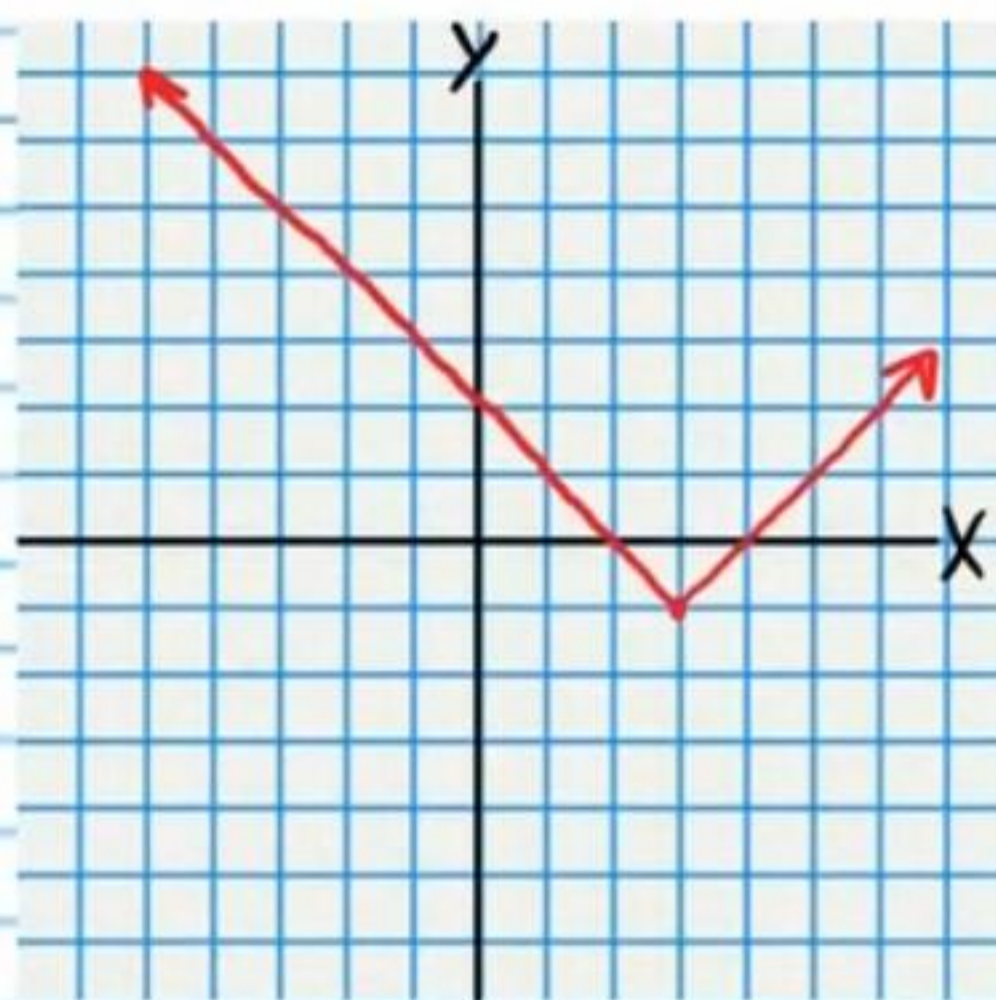
Parent Equation: $y = x^2$



⑦ $y+1 = |x-3|$

Parent Relation: Absolute Value

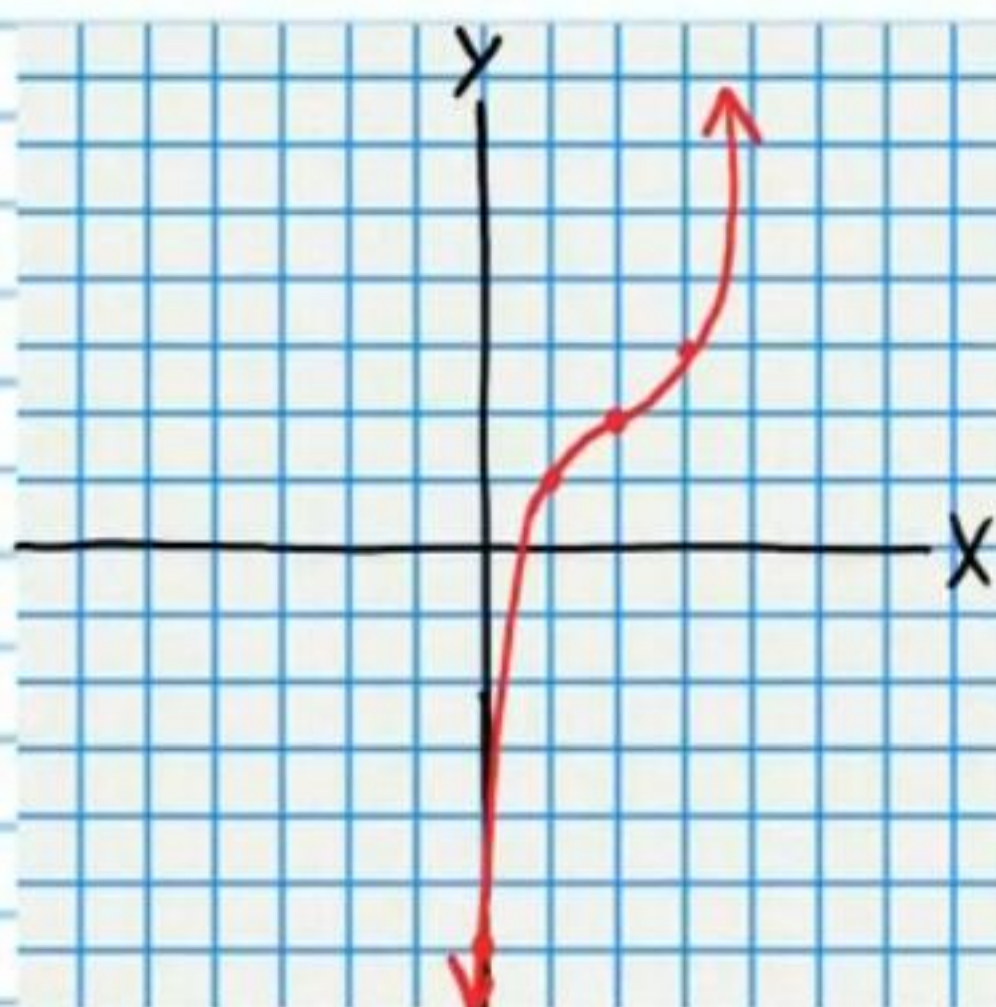
Parent Equation: $y = |x|$



⑧ $y-2 = (x-2)^3$

Parent Relation: Cubic

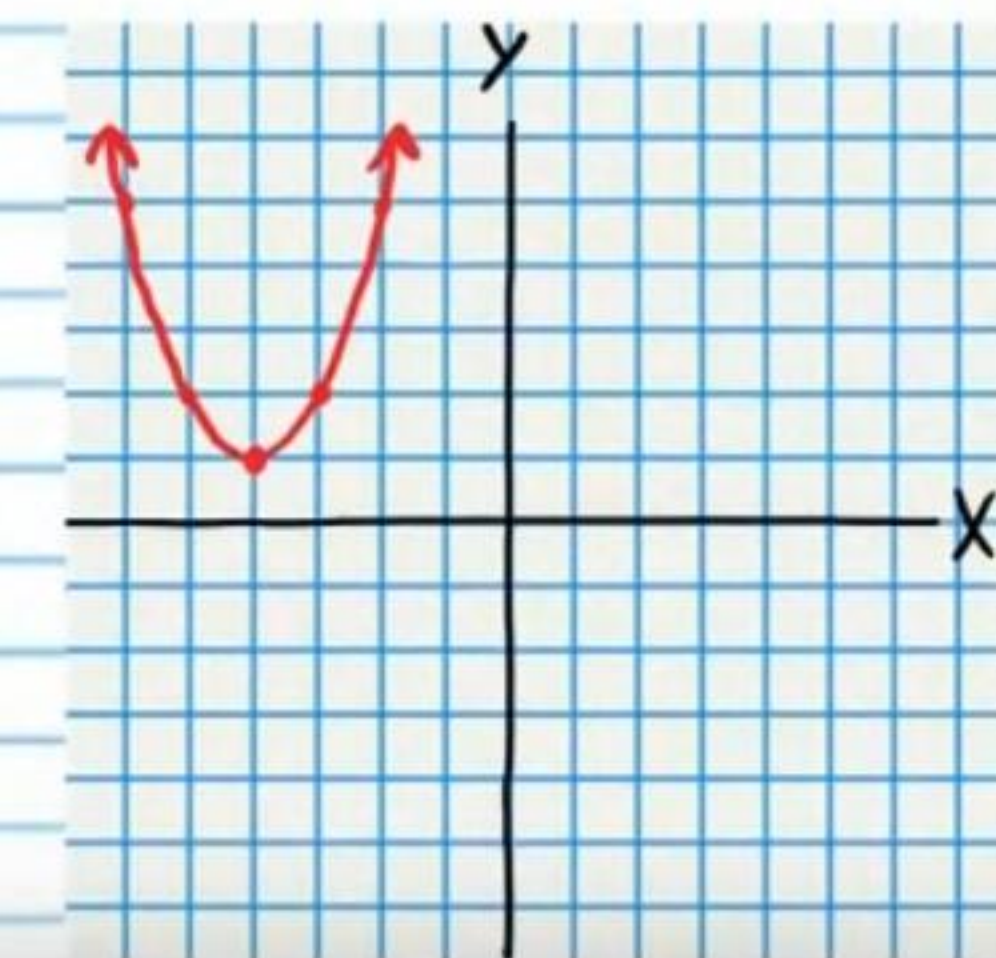
Parent Equation: $y = x^3$



⑨ $(x+4)^2 = y-1$

Parent Relation: Quadratic

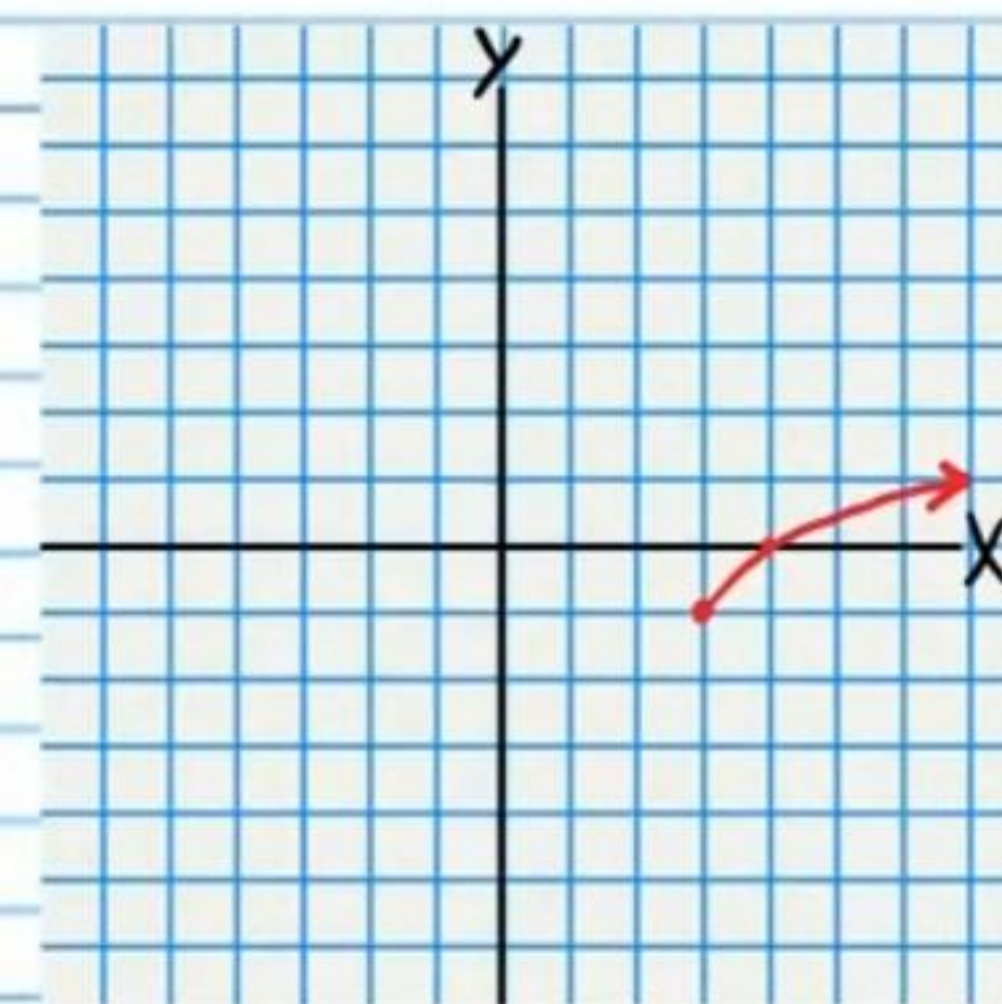
Parent Equation: $y = x^2$



⑩ $\sqrt{x-3} = y+1$

Parent Relation: Square Root

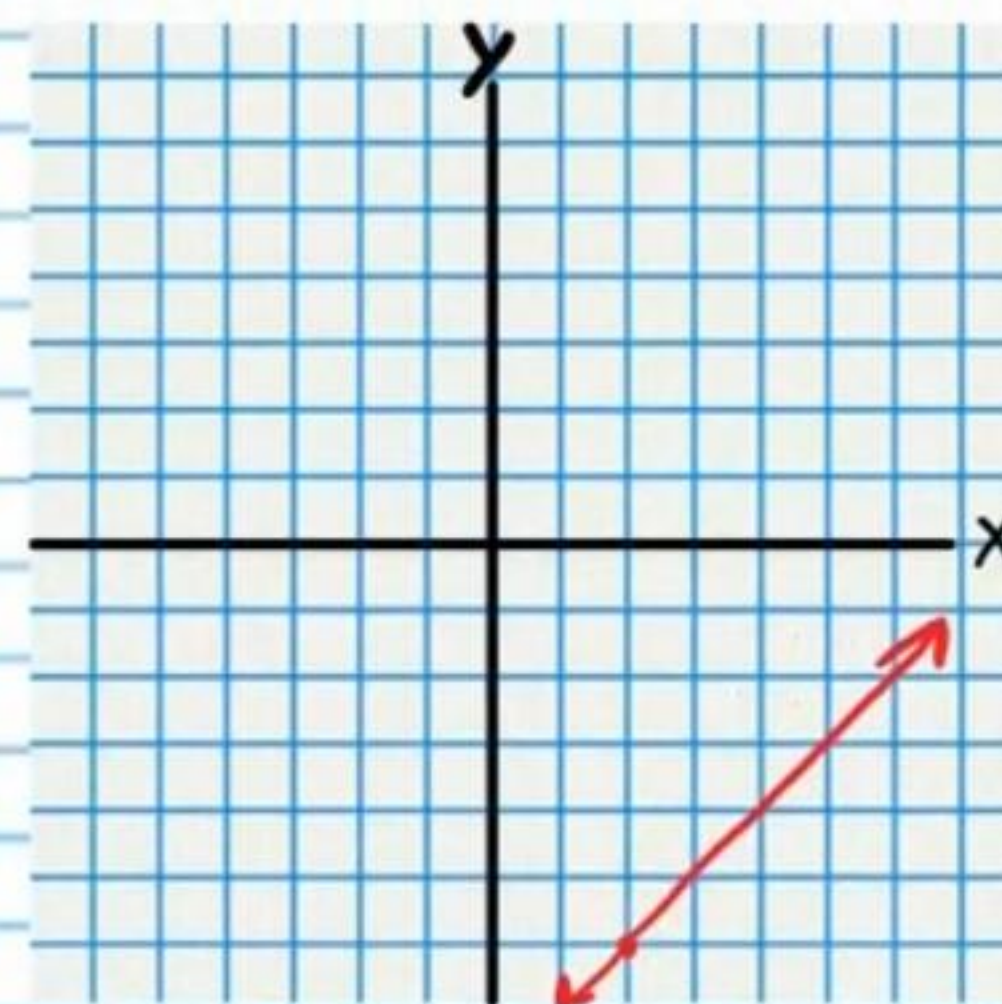
Parent Equation: $y = \sqrt{x}$



⑪ $y+6 = x-2$

Parent Relation: Linear

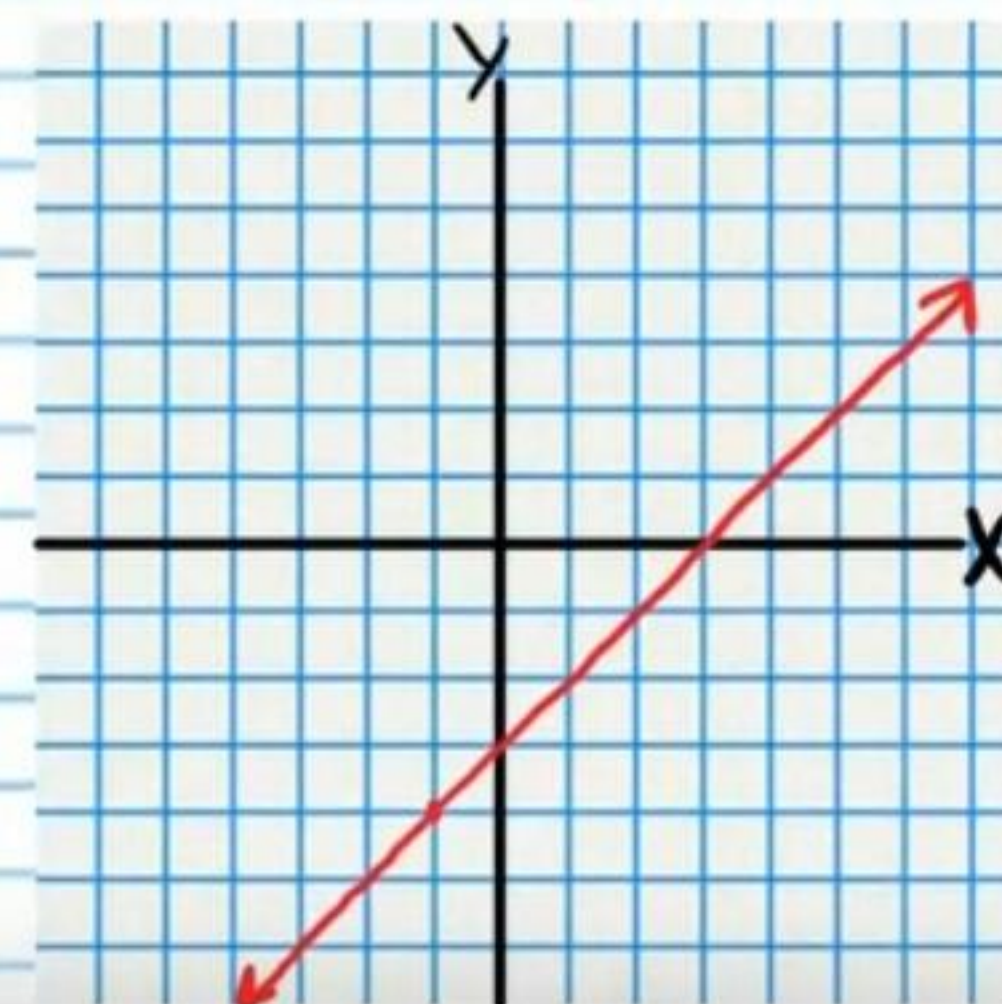
Parent Equation: $y = x$



⑫ $x+1 = y+4$

Parent Relation: Linear

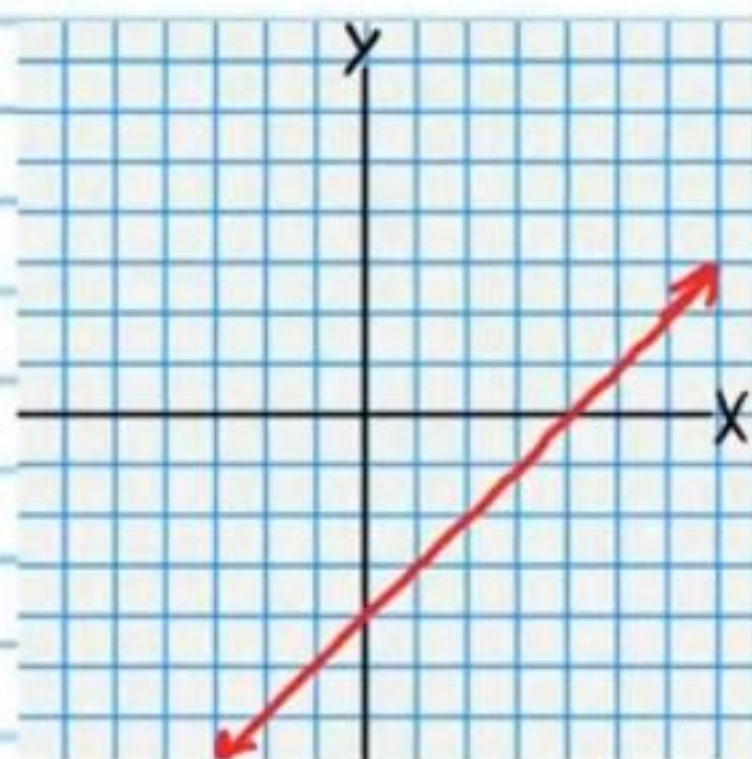
Parent Equation: $y = x$



⑬ $y - 2 = x - 6$

Parent Relation: Linear

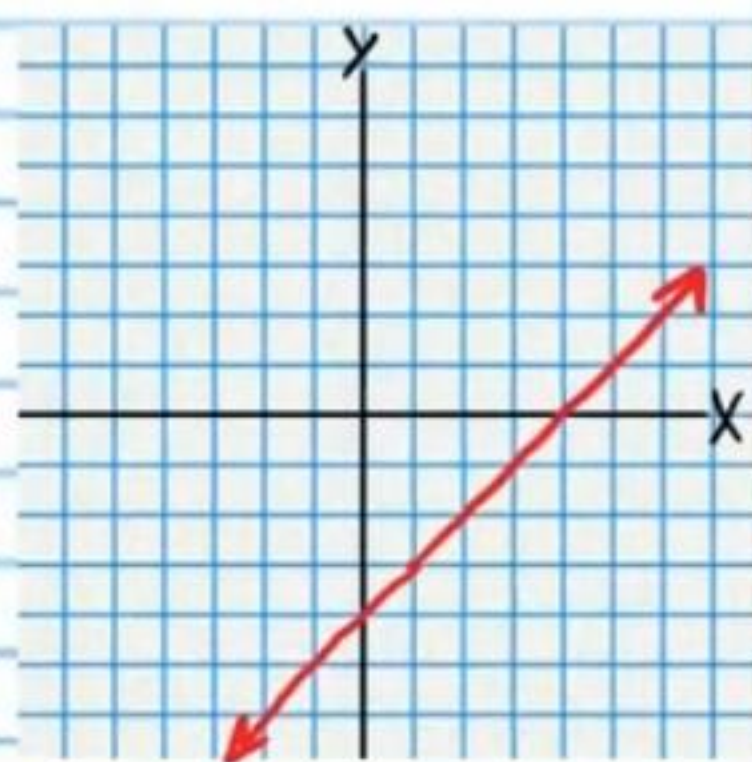
Parent Equation: $y = x$



⑭ $x - 1 = y + 3$

Parent Relation: Linear

Parent Equation: $y = x$



Reflections

Reflecting a Graph Across the X-Axis

If y is changed to $-y$ in an equation, then its corresponding graph is reflected across the x -axis.

Reflecting a Graph Across the Y-Axis

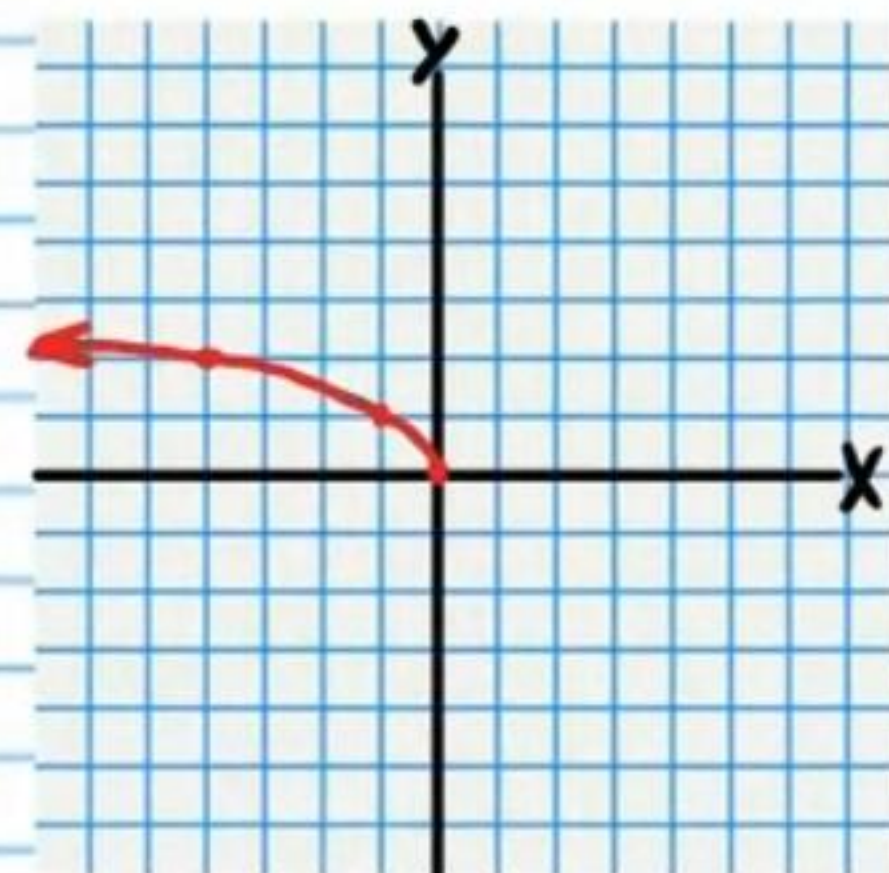
If x is changed to $-x$ in an equation, then its corresponding graph is reflected across the y -axis.

Write the parent relation and the parent equation of each relation below and use this information to graph the modified equation.

⑮ $y = \sqrt{-x}$

Parent Relation: Square Root

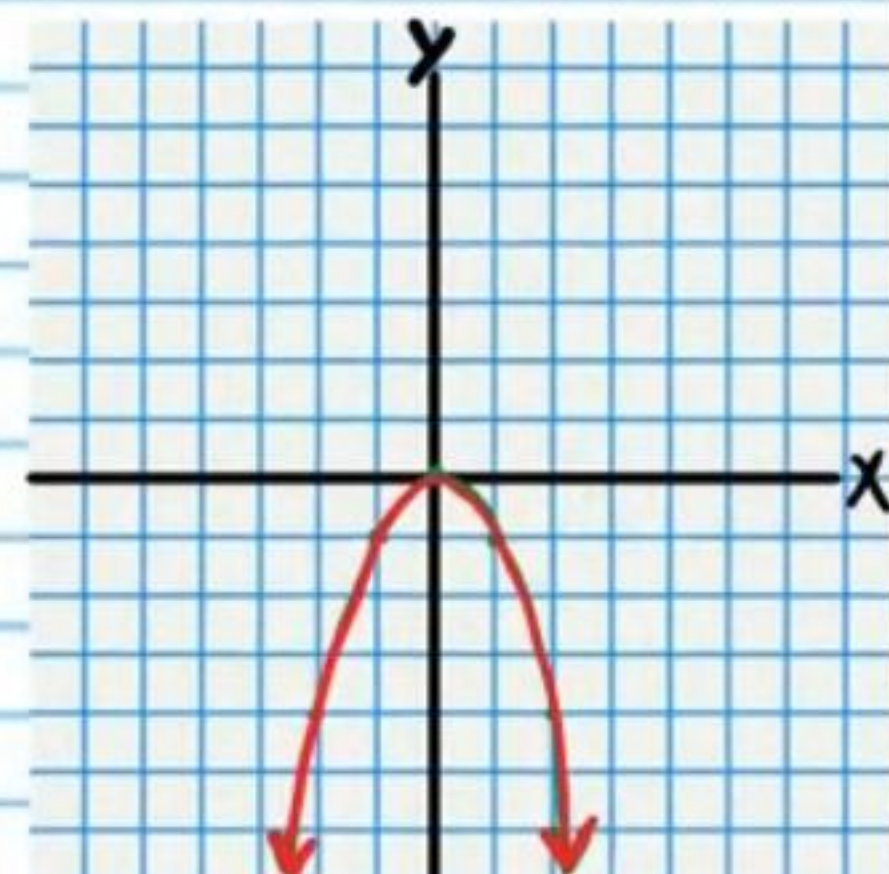
Parent Equation: $y = \sqrt{x}$



⑯ $-y = x^2$

Parent Relation: Quadratic

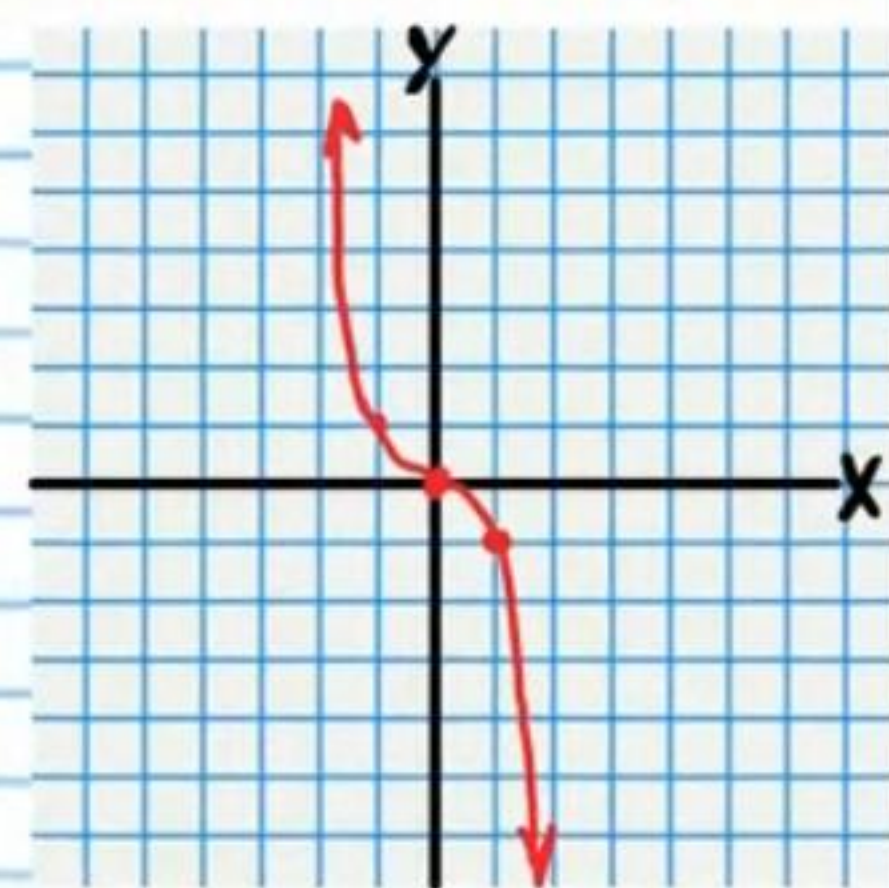
Parent Equation: $y = x^2$



⑰ $y = (-x)^3$

Parent Relation: Cubic

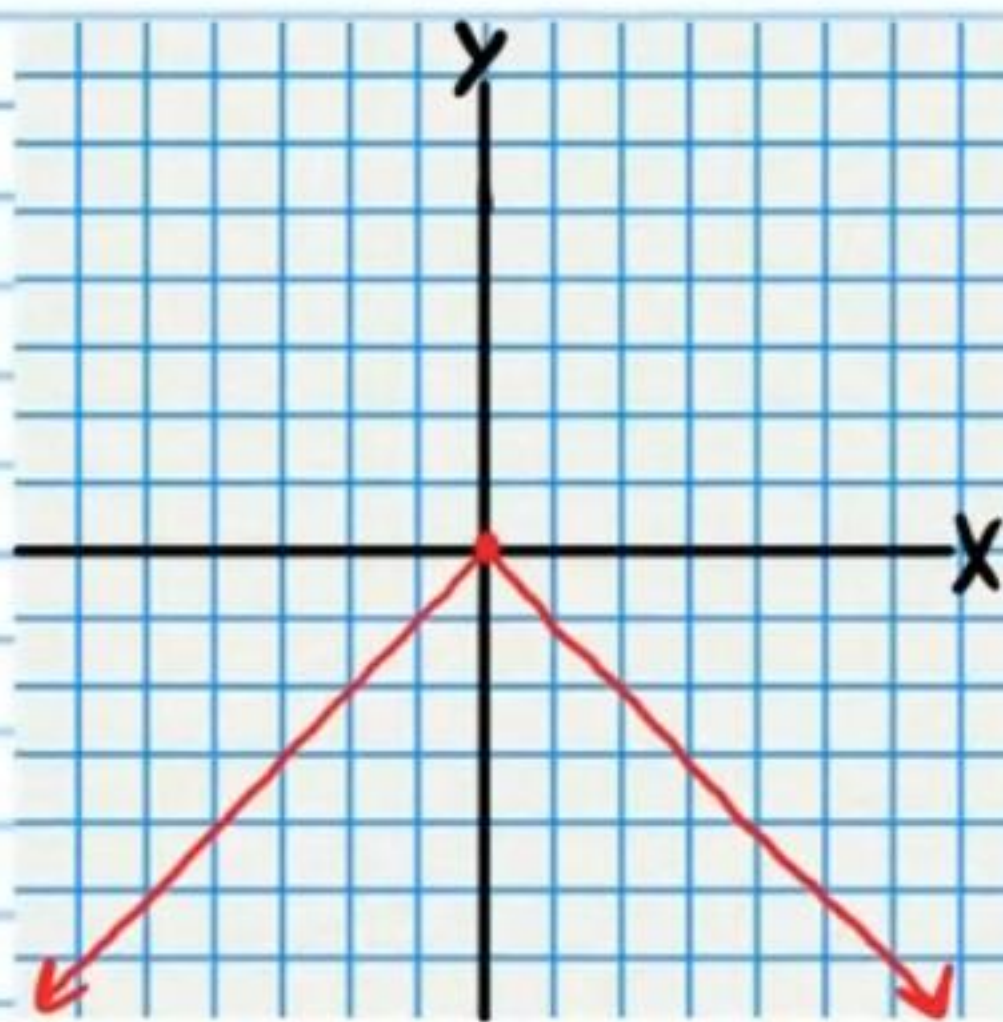
Parent Equation: $y = x^3$



⑮ $-y = |x|$

Parent Relation: Absolute Value

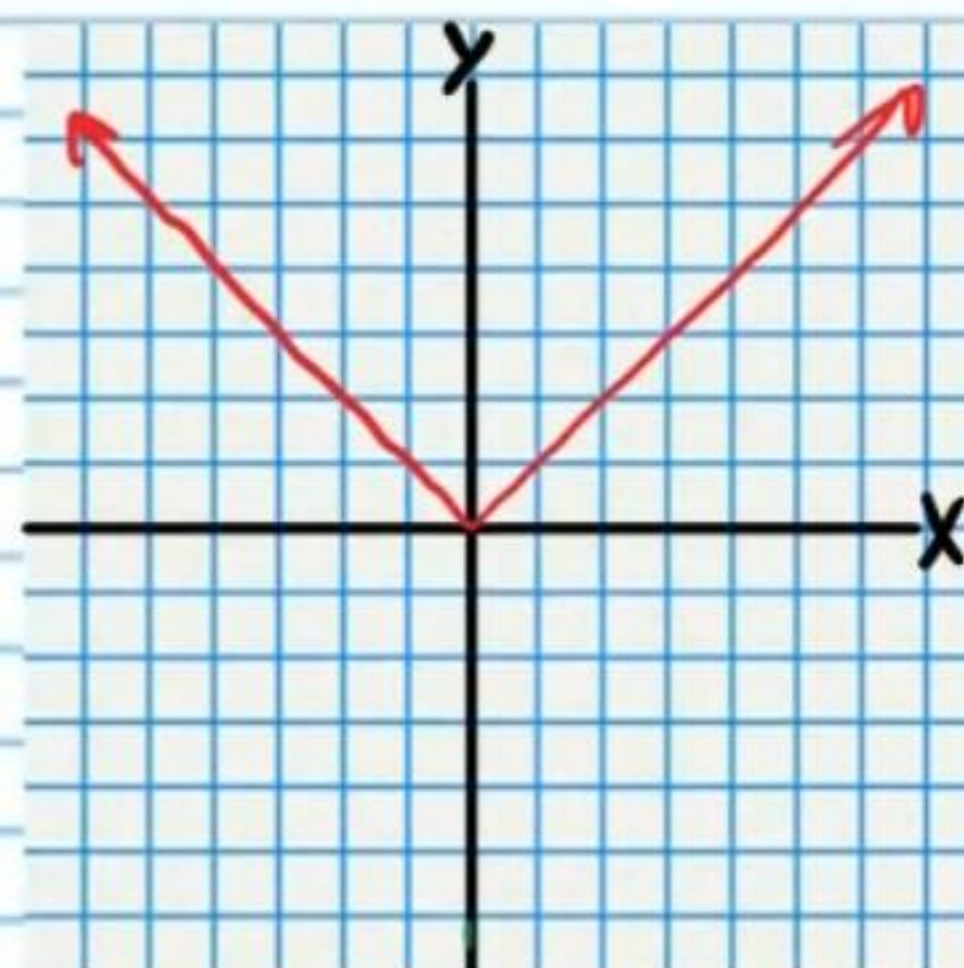
Parent Equation: $y = |x|$



★ ⑯ $y = |-x|$

Parent Relation: Absolute Value

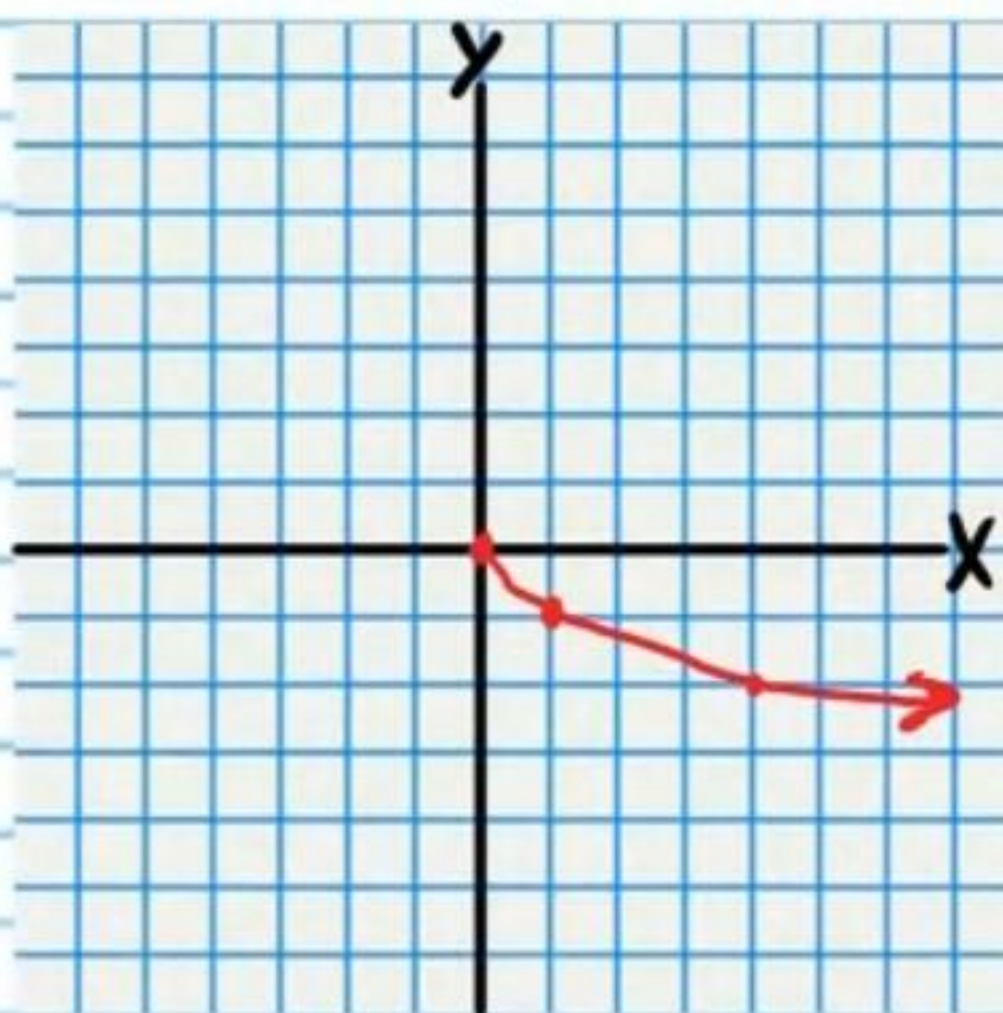
Parent Equation: $y = |x|$



⑰ $-y = \sqrt{x}$

Parent Relation: Square Root

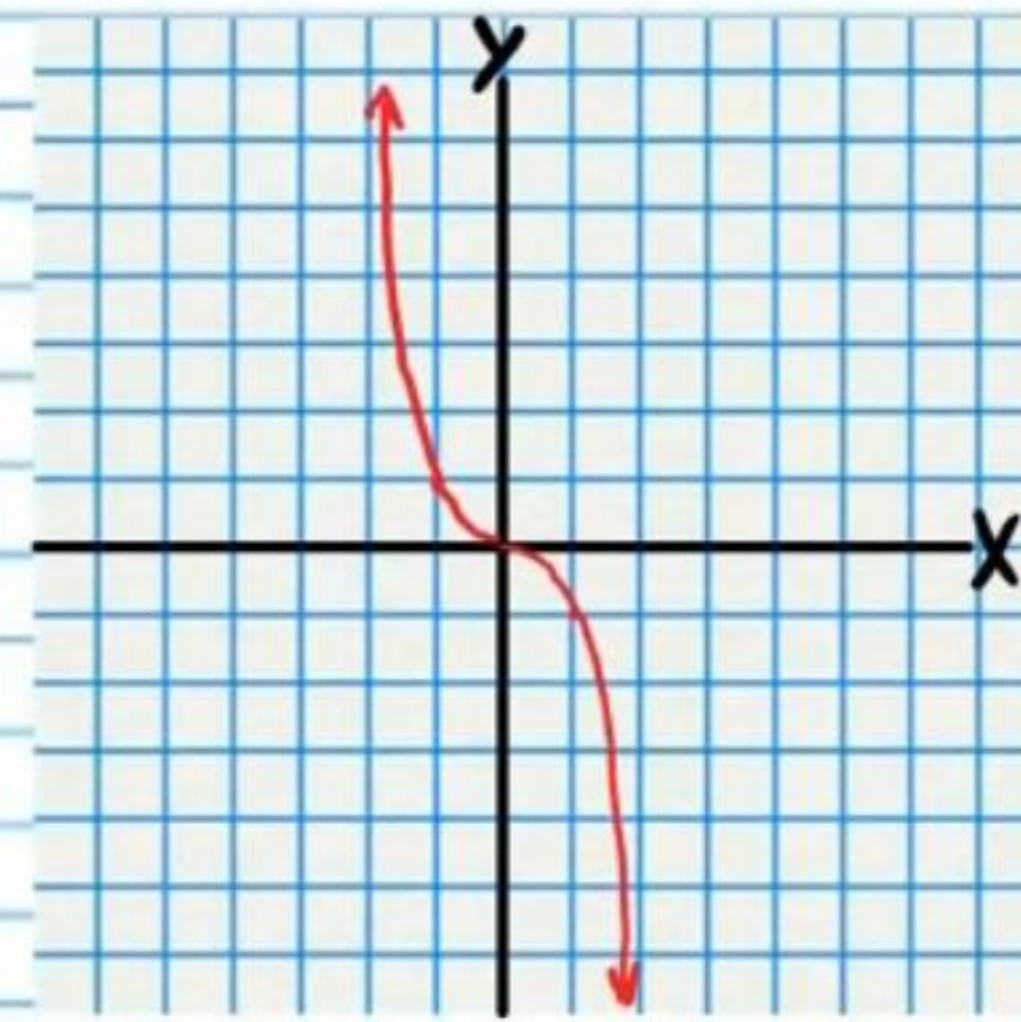
Parent Equation: $y = \sqrt{x}$



⑱ $-y = x^3$

Parent Relation: Cubic

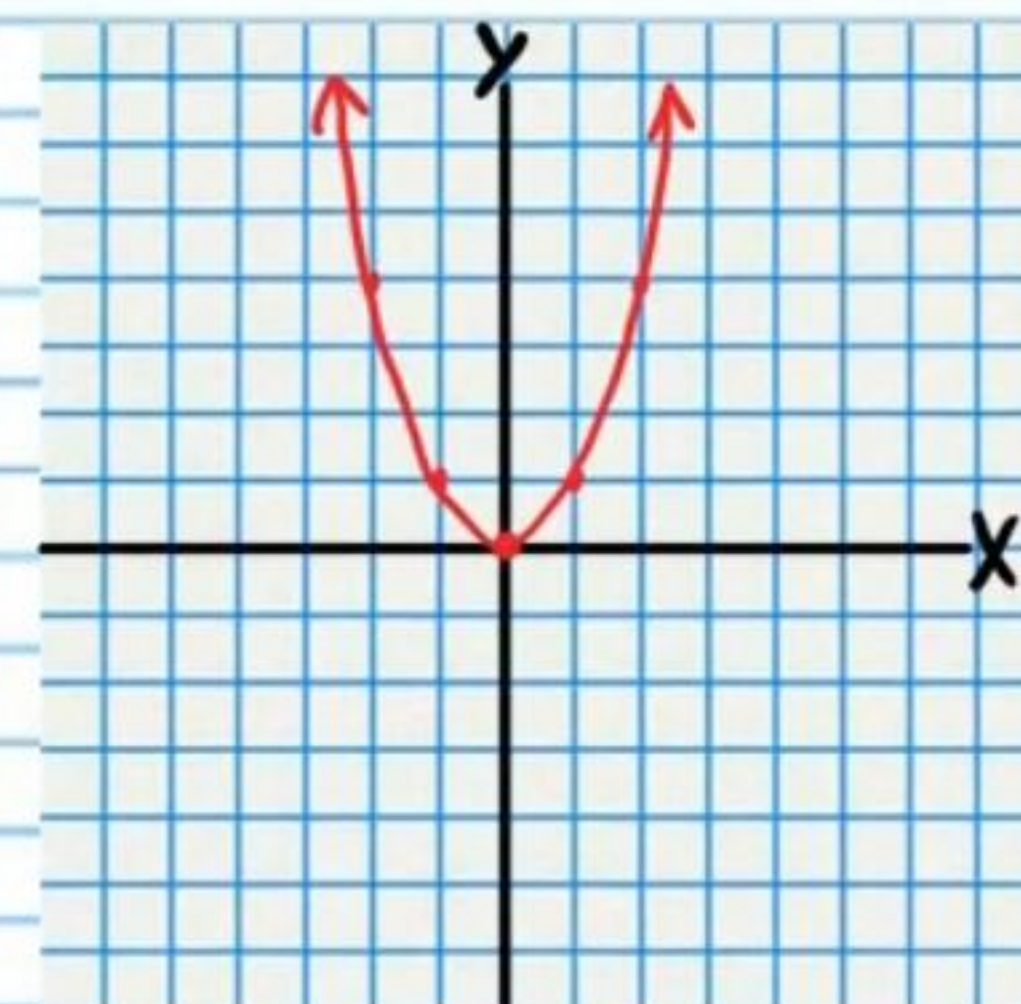
Parent Equation: $y = x^3$



★ ⑲ $y = (-x)^2$

Parent Relation: Quadratic

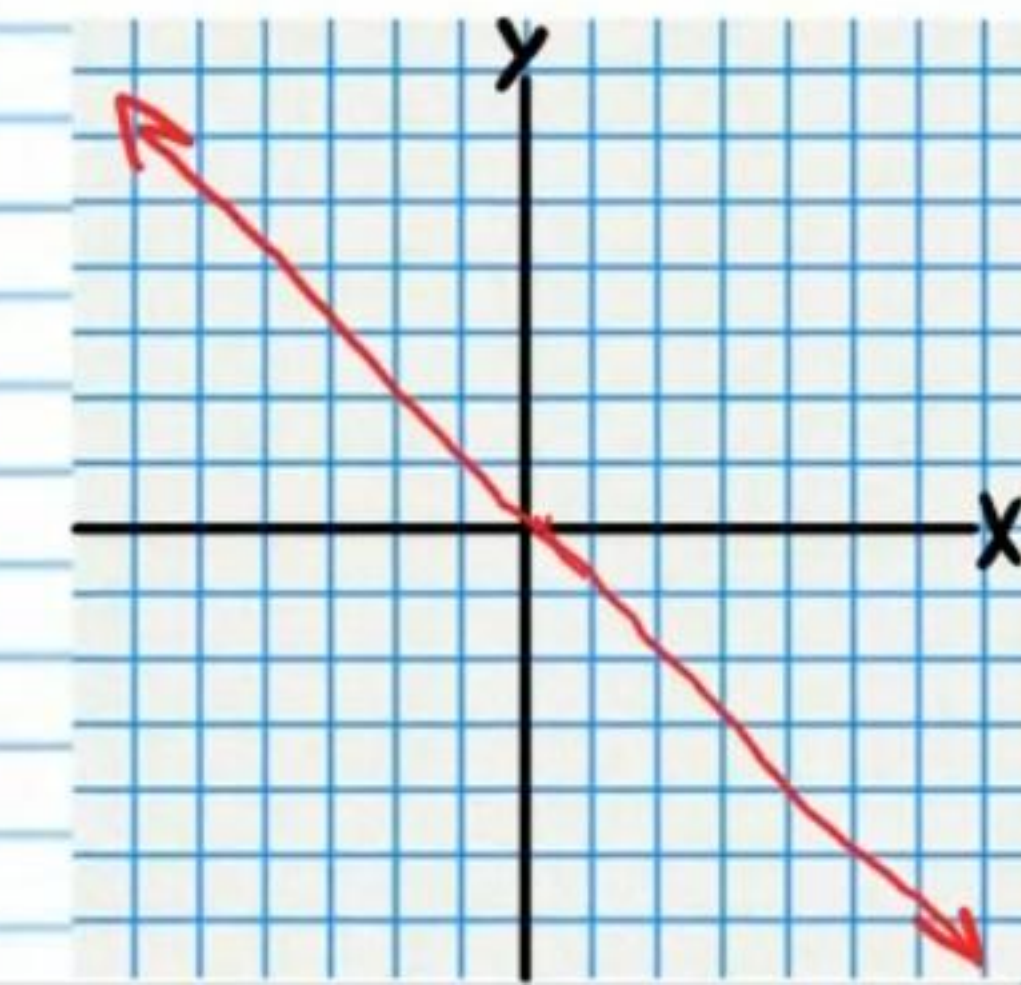
Parent Equation: $y = x^2$



⑳ $y = -x$

Parent Relation: Linear

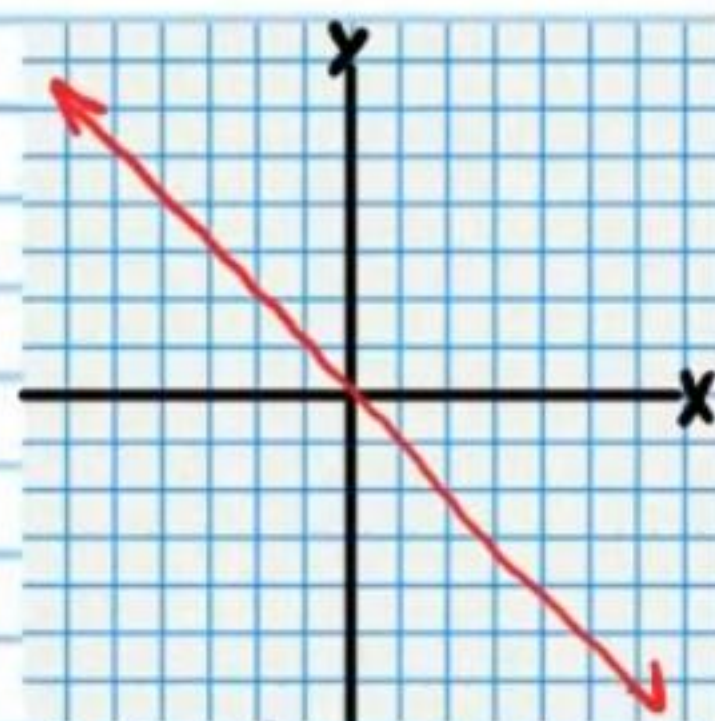
Parent Equation: $y = x$



②④ $-y = x$

Parent Relation: Linear

Parent Equation: $y = x$



Dilations (Compressing and Stretching)

Dilating a Graph Vertically

If y is **multiplied** by a number in an equation, then its corresponding graph is **compressed vertically**. All of the y -values of the graph of the previous equation will be **divided** by the number and each corresponding x -value will remain the same.

If y is **divided** by a number in an equation, then its corresponding graph is **stretched vertically**. All of the y -values of the graph of the previous equation will be **multiplied** by the number and each corresponding x -value will remain the same.

Dilating a Graph Horizontally

If x is **multiplied** by a number in an equation, then its corresponding graph is **compressed horizontally**. All of the x -values of the graph of the previous equation will be **divided** by the number and each corresponding y -value will remain the same.

If x is **divided** by a number in an equation, then its corresponding graph is **stretched horizontally**. All of the x -values of the graph of the previous equation will be **multiplied** by the number and each corresponding y -value will remain the same.

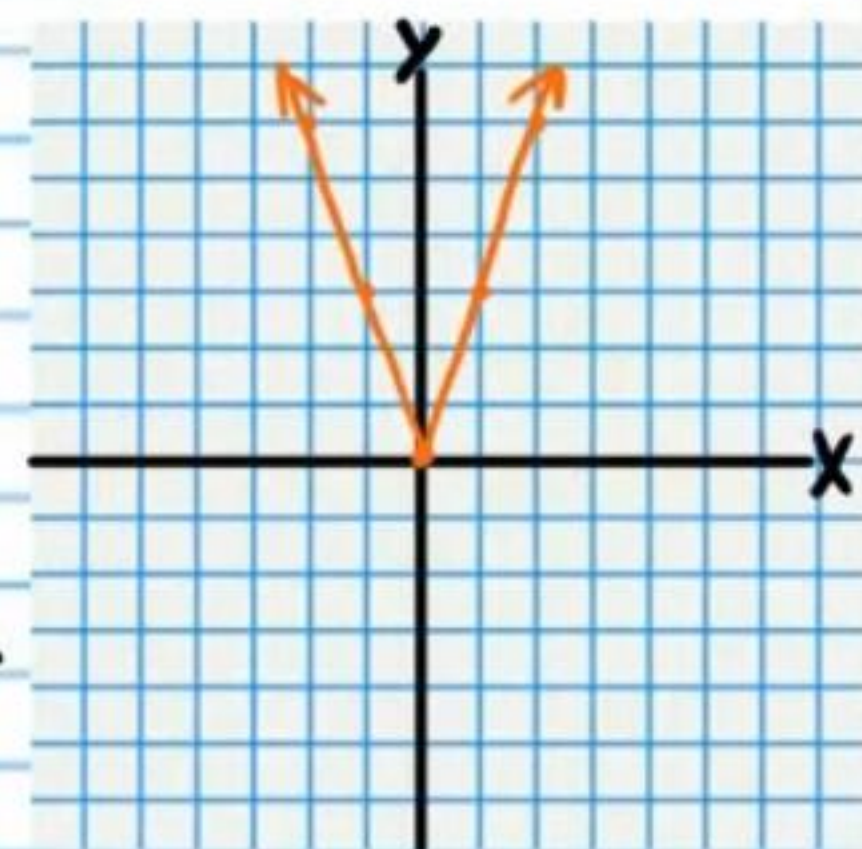
Write the parent relation and the parent equation of each relation below and use this information to graph the modified equation. Then, write the type of Dilation involved.

②⑤ $y = |3x|$

Parent Relation: Absolute Value

Parent Equation: $y = |x|$

Dilation: Horizontal compression.

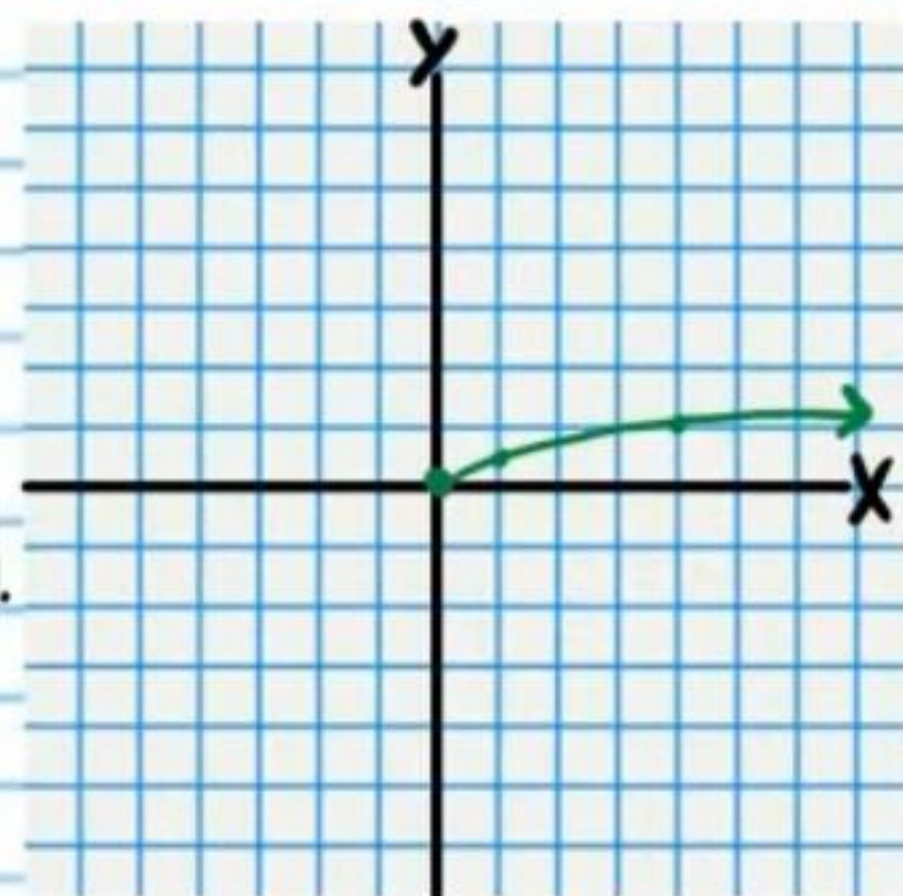


②⑥ $2y = \sqrt{x}$

Parent Relation: Square Root

Parent Equation: $y = \sqrt{x}$

Dilation: Vertical (compression).

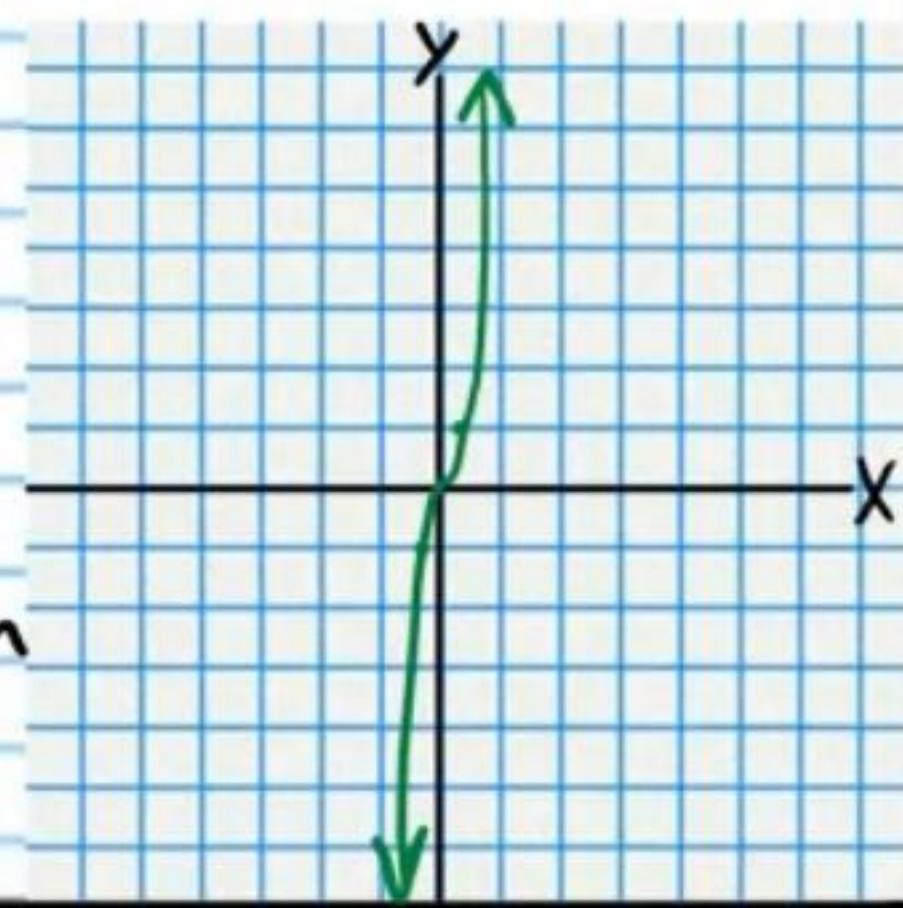


②⑦ $y = (4x)^3$

Parent Relation: Cubic

Parent Equation: $y = x^3$

Dilation: Horizontal compression

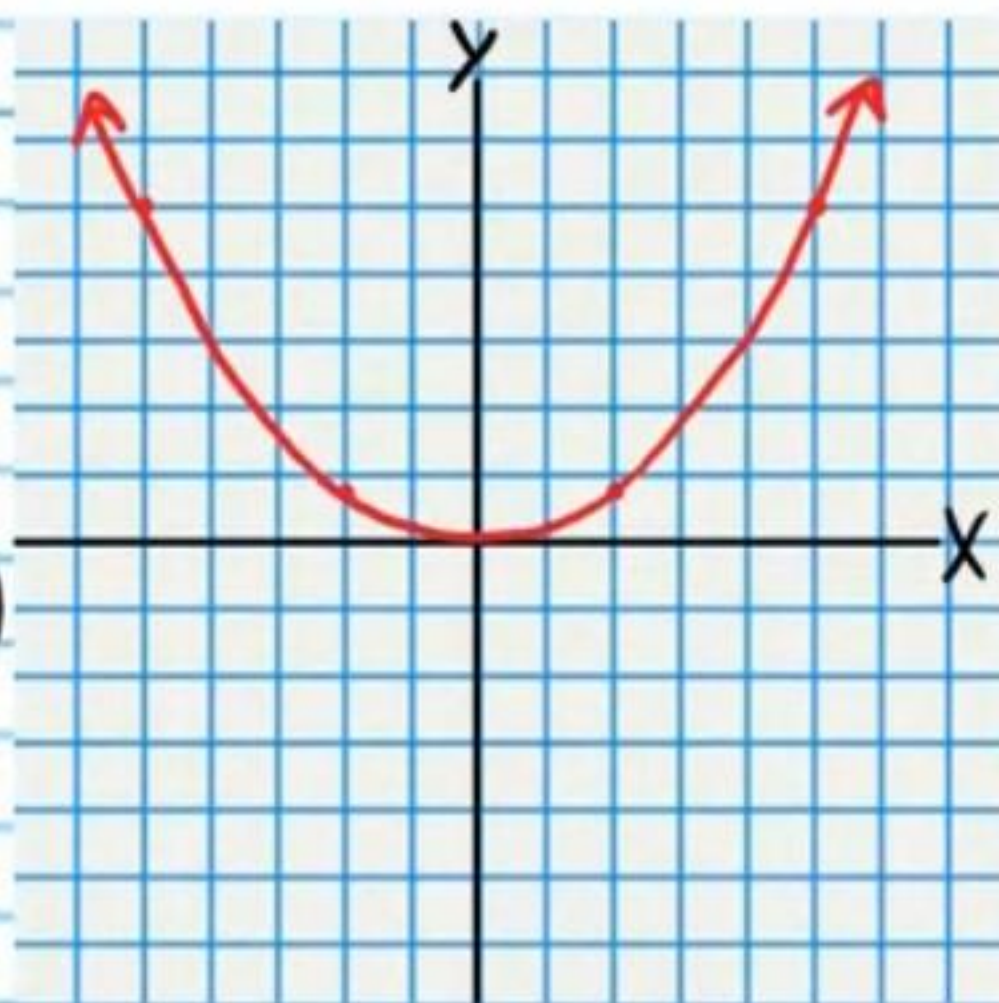


②⑧ $5y = x^2$

Parent Relation: Quadratic

Parent Equation: $y = x^2$

Dilation: Vertical (compression)

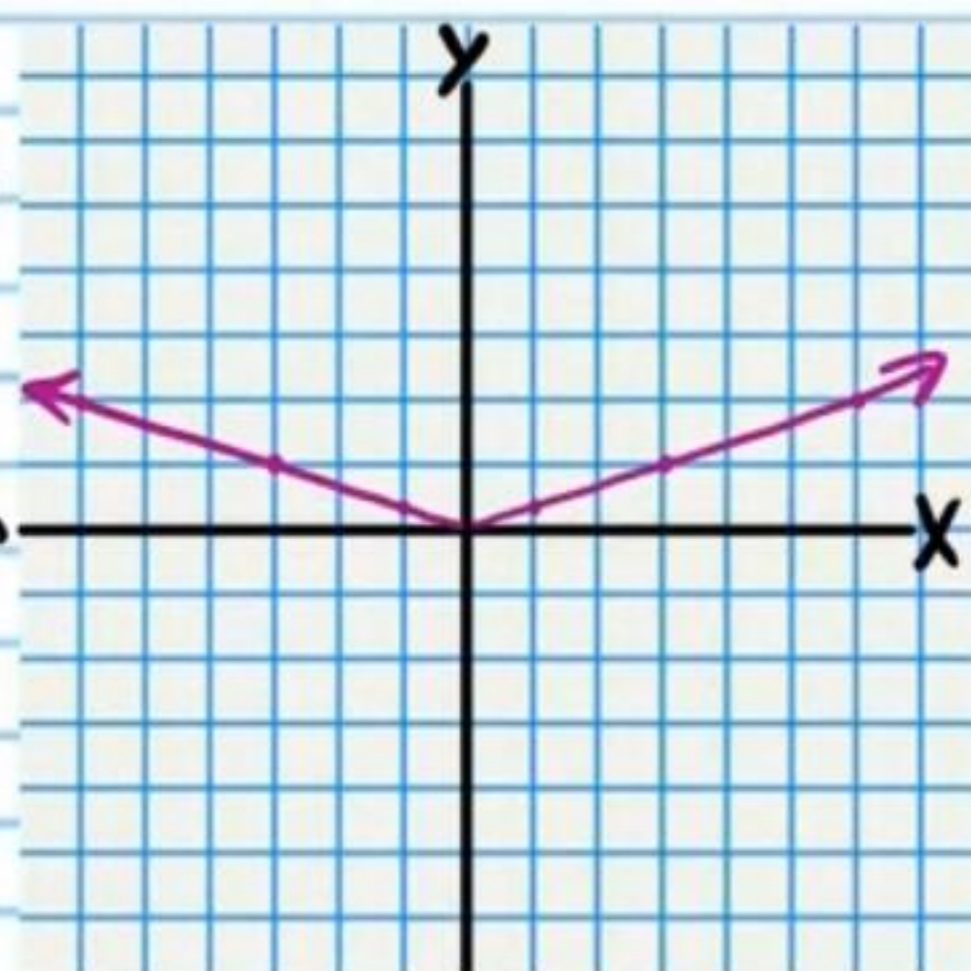


②⑨ $3y = |x|$

Parent Relation: Absolute Value

Parent Equation: $y = |x|$

Dilation: Vertical compression

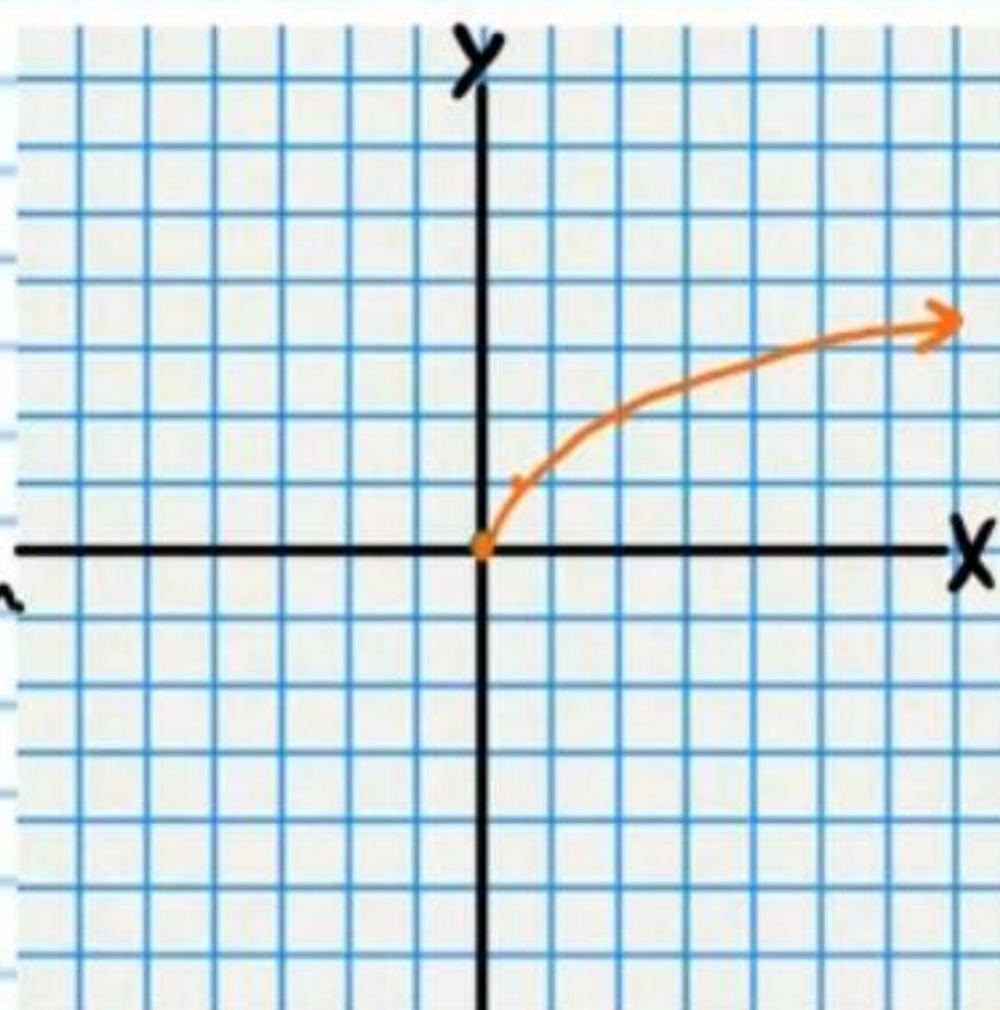


③⑩ $y = \sqrt{2x}$

Parent Relation: Square Root

Parent Equation: $y = \sqrt{x}$

Dilation: Horizontal compression

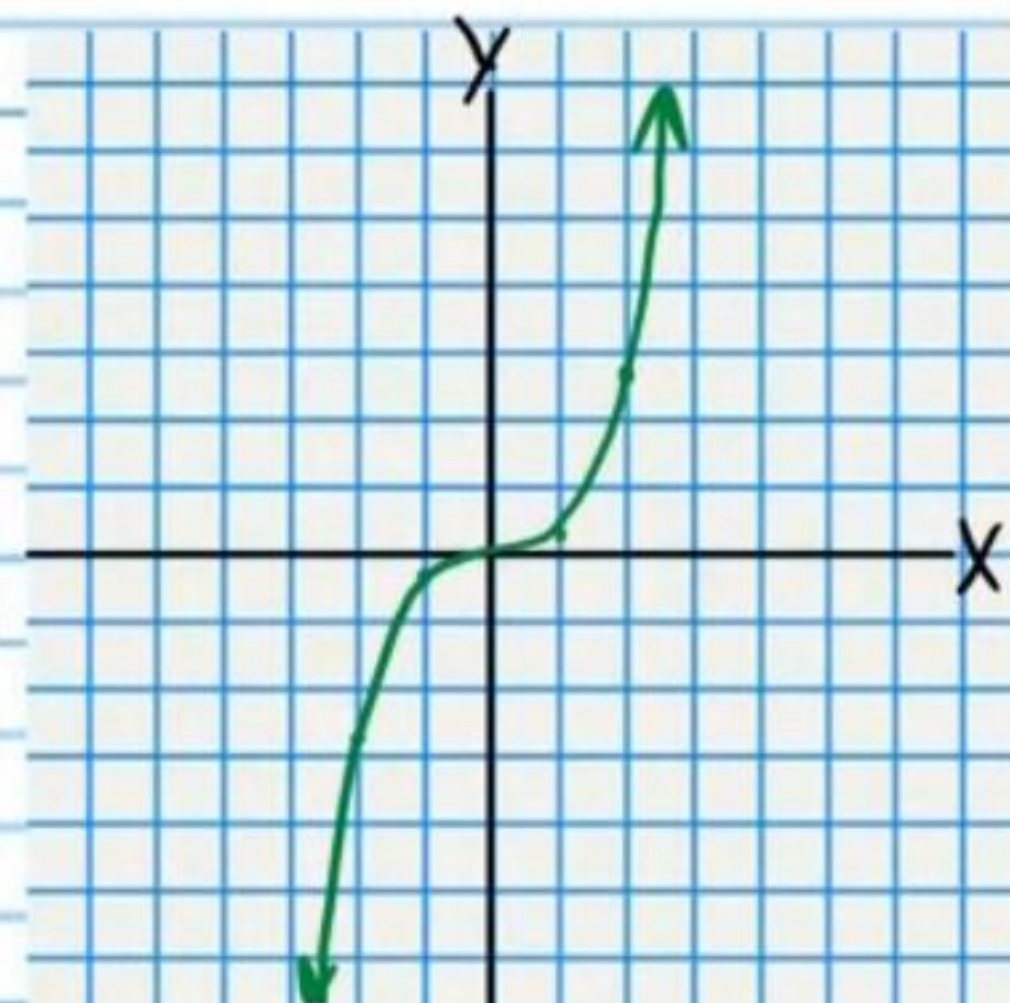


③① $3y = x^3$

Parent Relation: Cubic

Parent Equation: $y = x^3$

Dilation: Vertical compression

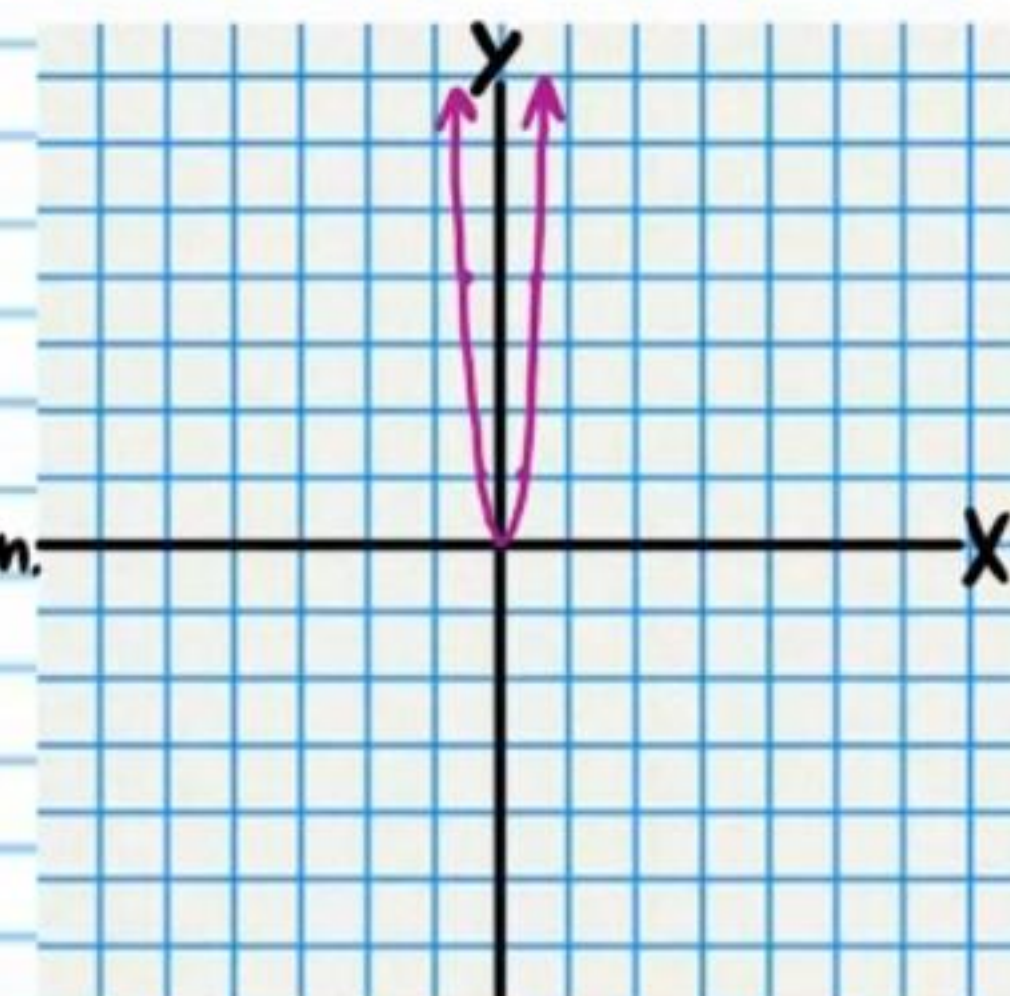


③② $y = (4x)^2$

Parent Relation: Quadratic

Parent Equation: $y = x^2$

Dilation: Horizontal compression

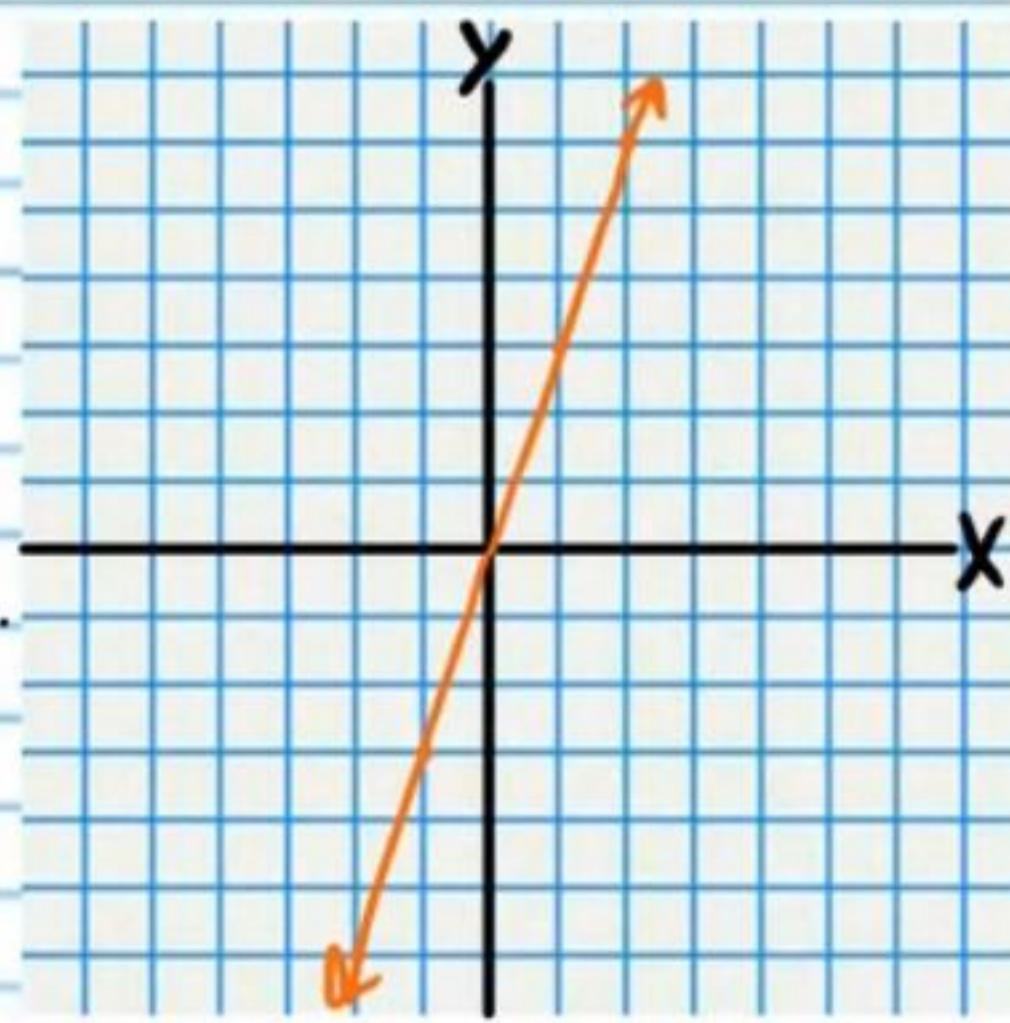


③③ $y = 3x$

Parent Relation: Linear

Parent Equation: $y = x$

Dilation: Horizontal compression

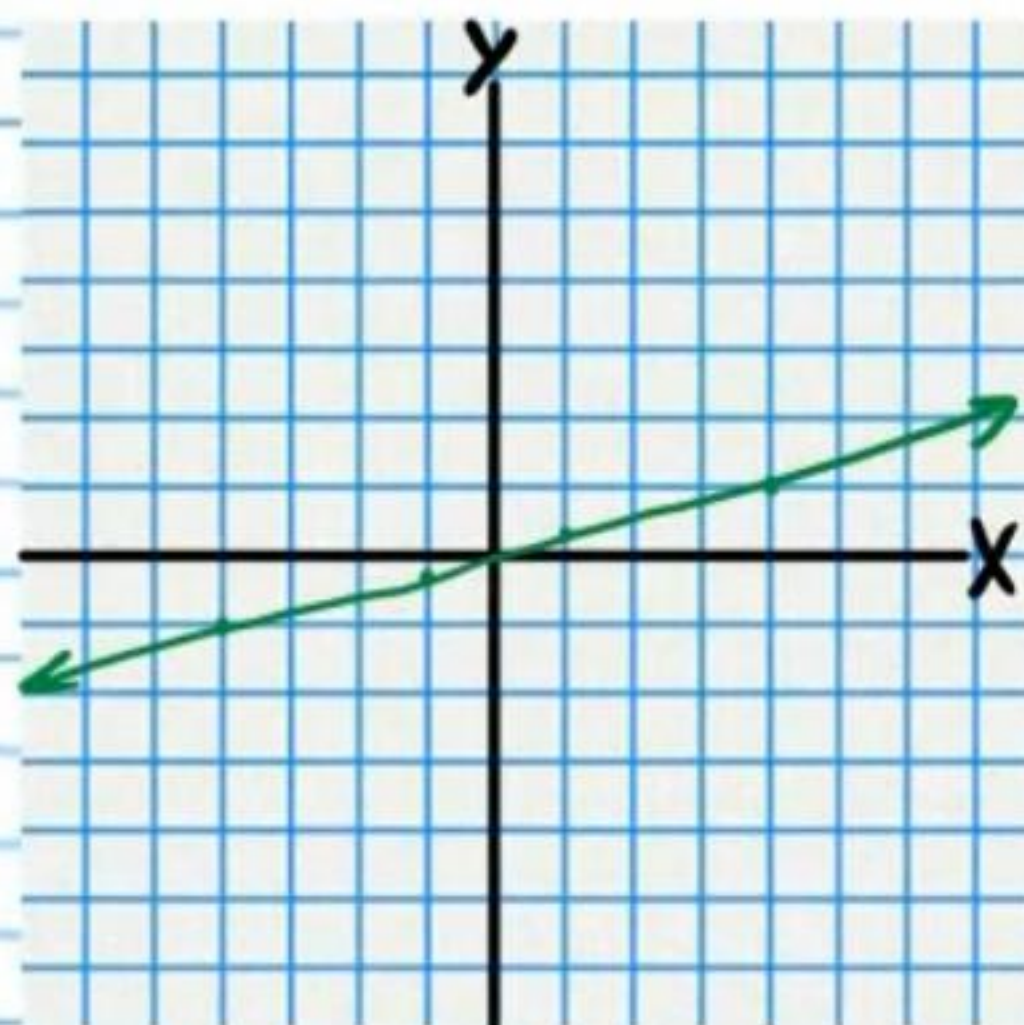


③④ $4y = x$

Parent Relation: Linear

Parent Equation: $y = x$

Dilation: Vertical Compression

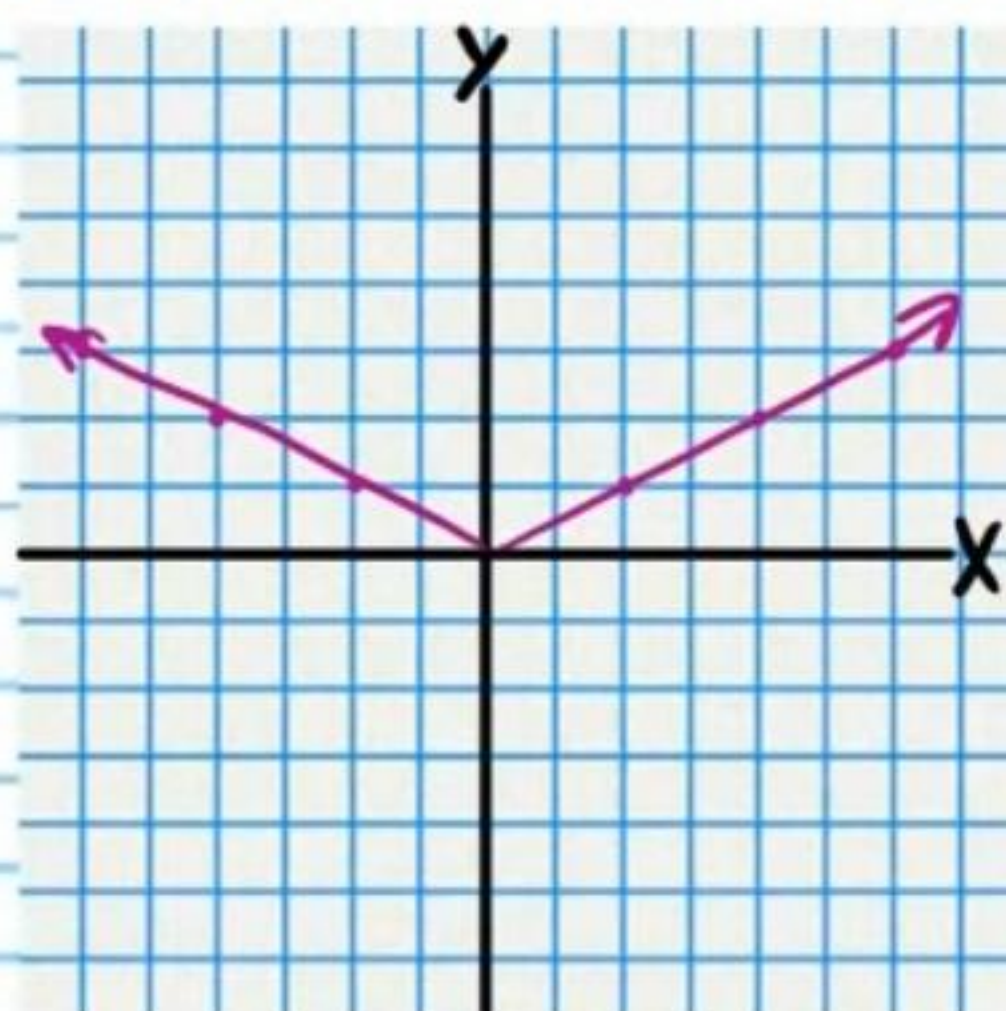


③⑤ $y = \left|\frac{x}{2}\right|$

Parent Relation: Absolute Value

Parent Equation: $y = |x|$

Dilation: Horizontal Stretch

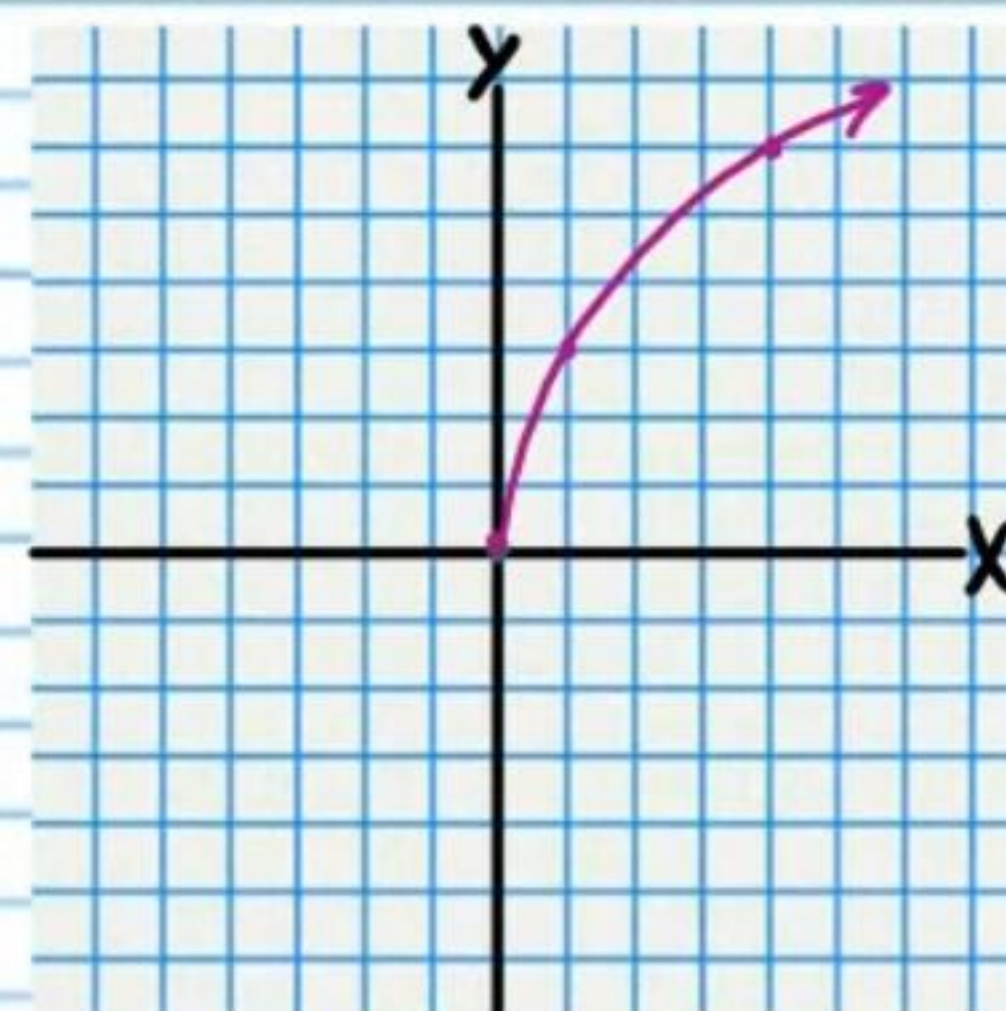


③⑥ $\frac{y}{3} = \sqrt{x}$

Parent Relation: Square Root

Parent Equation: $y = \sqrt{x}$

Dilation: Vertical Stretch

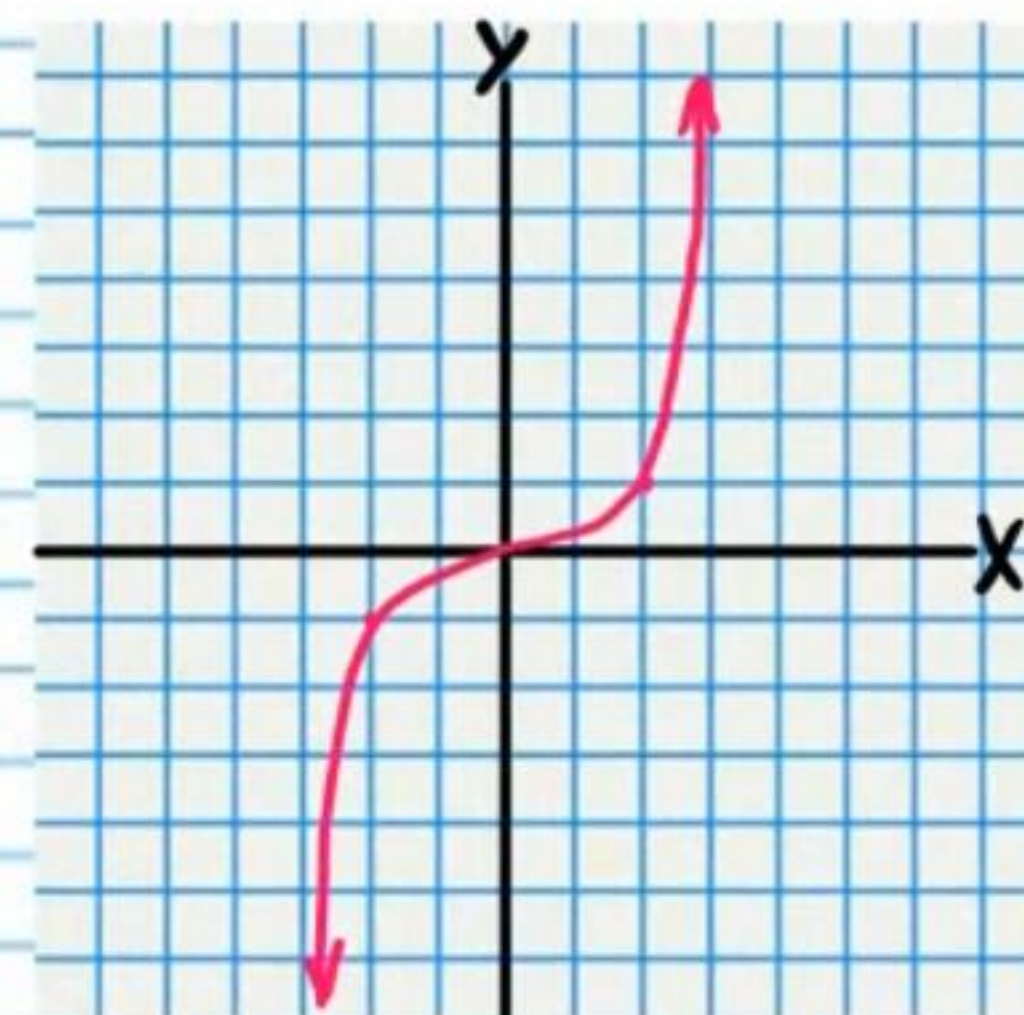


③⑦ $y = \left(\frac{x}{2}\right)^3$

Parent Relation: Cubic

Parent Equation: $y = x^3$

Dilation: Horizontal Stretch

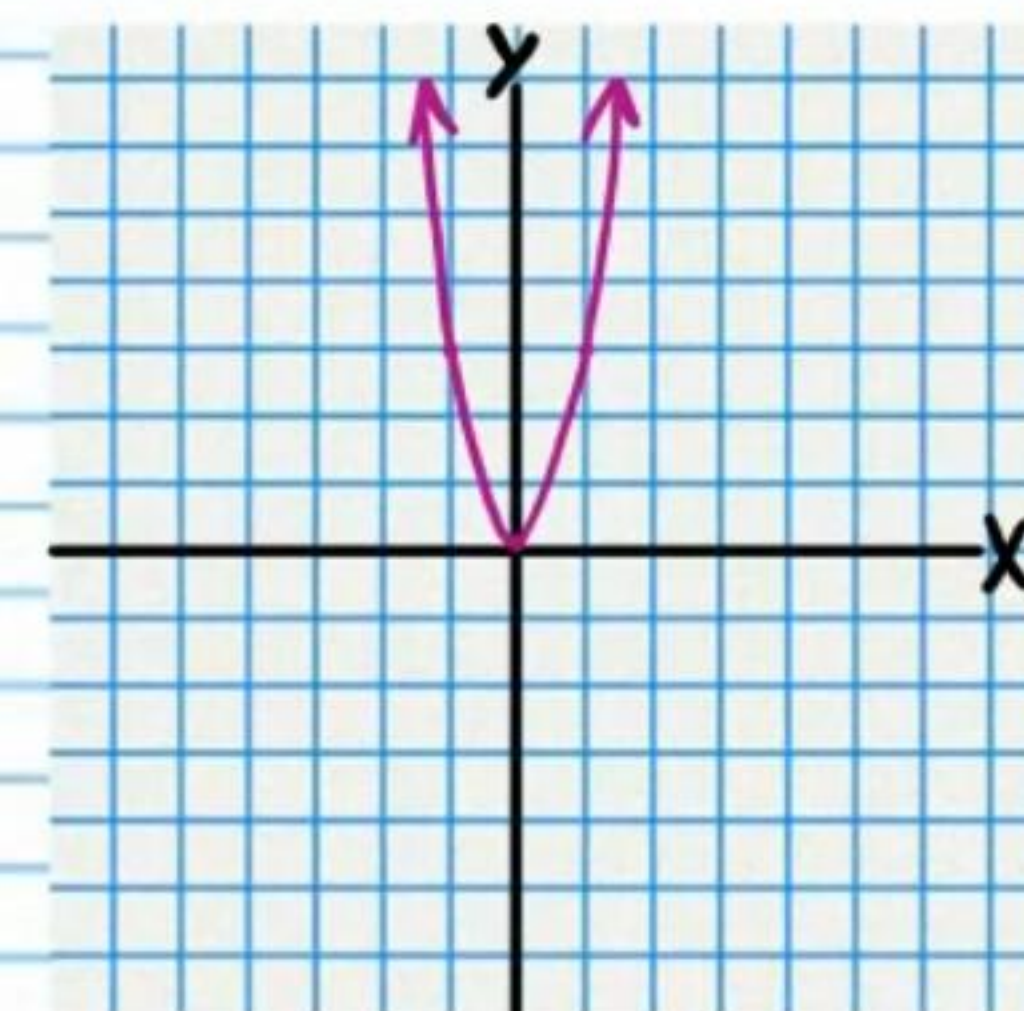


③⑧ $\frac{y}{3} = x^2$

Parent Relation: Quadratic

Parent Equation: $y = x^2$

Dilation: Vertical Stretch

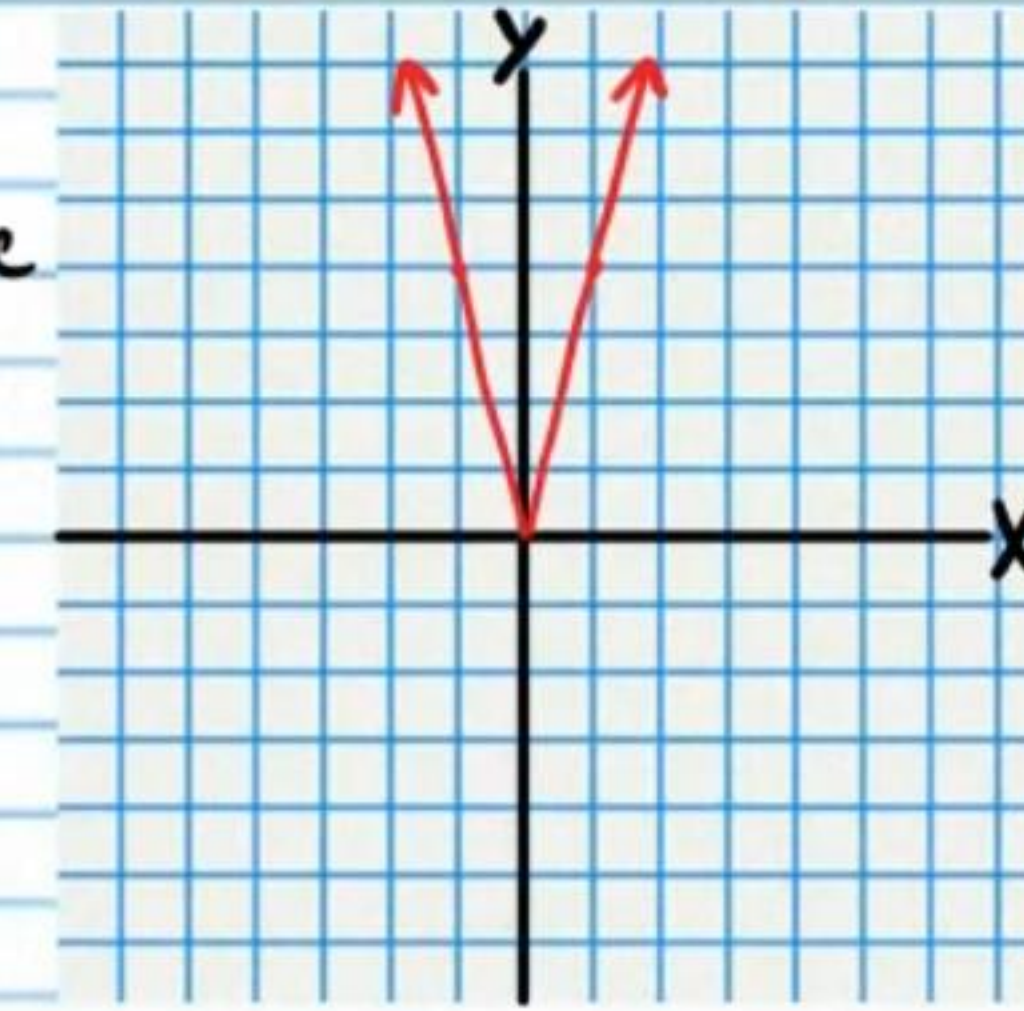


③⑨ $\frac{y}{4} = |x|$

Parent Relation: Absolute Value

Parent Equation: $y = |x|$

Dilation: Vertical Stretch

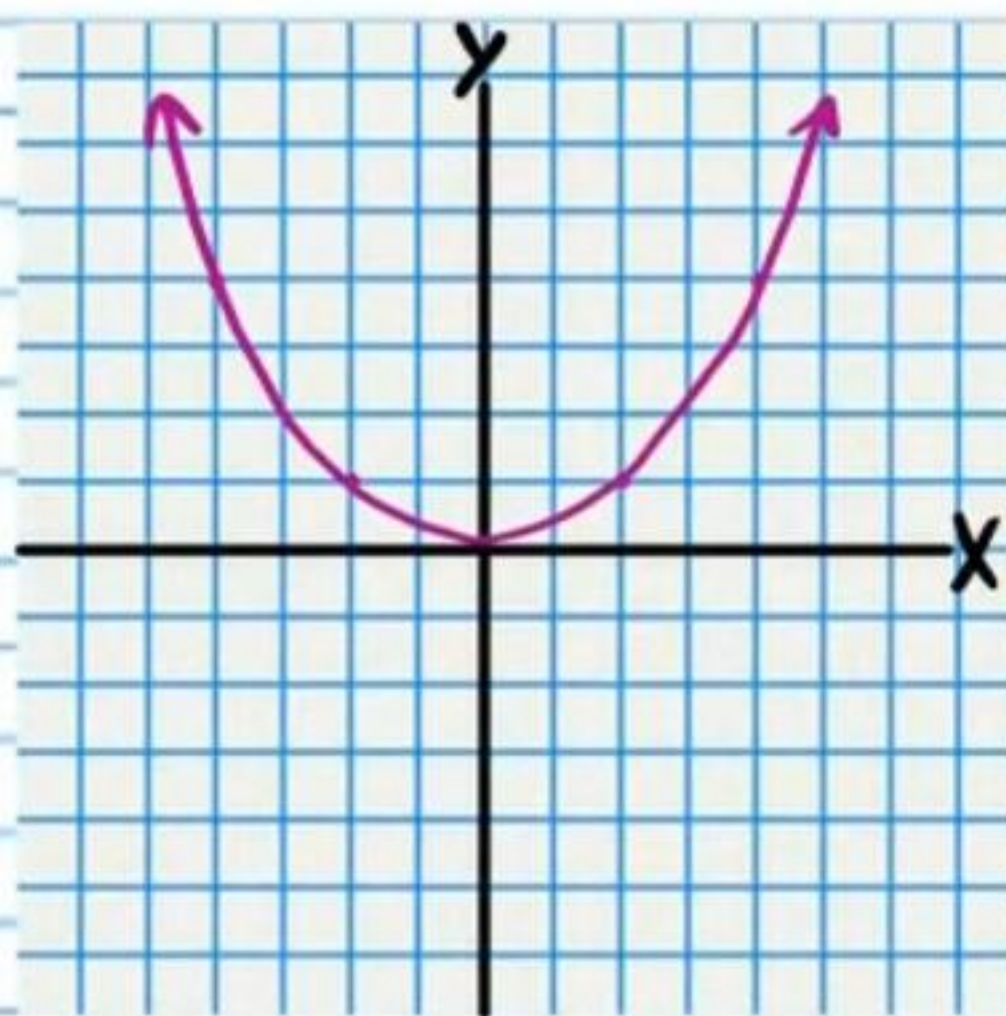


④① $y = \left(\frac{x}{2}\right)^2$

Parent Relation: Quadratic

Parent Equation: $y = x^2$

Dilation: Horizontal Stretch.

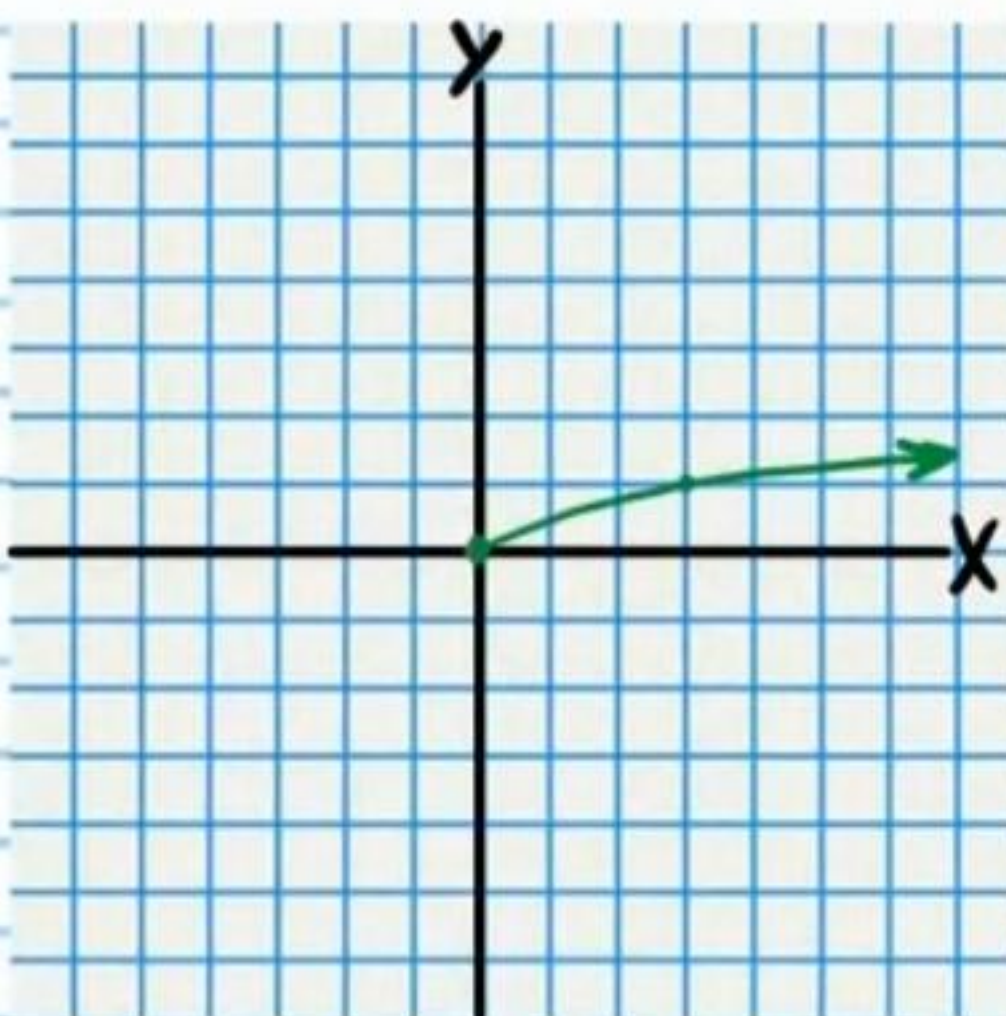


④② $y = \sqrt{\frac{x}{3}}$

Parent Relation: Square Root

Parent Equation: $y = \sqrt{x}$

Dilation: Horizontal Stretch



④③ $\frac{y}{2} = x^3$

Parent Relation: Cubic

Parent Equation: $y = x^3$

Dilation: Vertical Stretch.

